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A theory of measurement in diagnosis from first principles

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Abstract

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Reiter and de Kleer have independently developed a theory of diagnosis from first principles. Reiter's approach to computing all diagnoses for a given faulty system is based upon the computation of all minimal hitting sets for the collection of conflict sets for (SD,COMPONENTS,OBS). Unfortunately, his theory does not include a theory of measurement. De Kleer and Williams have developed GDE—general diagnostic engine. Their procedure computes all minimal conflict sets resulting from a measurement before discriminating the candidate space. However, they do not provide a formal justification for their theory.

We propose a general theory of measurement in diagnosis and provide a formal justification for our theory. Several novel contributions make up the central focus of this paper. First, this work provides an efficient incremental method for computing new diagnoses given a new measurement, based on the previous diagnoses predicting the opposite. Second, this work defines the concepts of conflict set resulting from a measurement, equivalence classes and homogeneous diagnoses as the basis of the method. Finally, this work leads to a procedure for computing all diagnoses and discriminating among competing diagnoses resulting from a measurement.

1. Introduction

Many researchers have developed systems and theories for diagnosis which use a "first principles approach" using a representation language generally based on first-order logic. For diagnosis from first principles, one should be given a description of some system—a physical device or real world setting of interest, say—together with an observation of the system's behaviour. If this observation conflicts with the way the system is meant to behave, one is

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confronted with a diagnostic problem. One must determine those system components which, when assumed to be functioning abnormally, will explain the discrepancy between the observed and correct system description, relying on its design or structure, together with observation(s) of the system behaviour. Notable examples of approaches to diagnostic reasoning from first principles are [3, 4, 7, 9, 14, 24, 26, 28].

Both Reiter [28] and de Kleer and Williams [7] use the concept of a conflict set as the basis of their methods. Davis [3] describes a technique called constraint suspension that provides a powerful tool for troubleshooting. As they point out, the diagnostic candidate may not be unique. Multiple competing diagnoses can arise from a given faulty system. The normal approach to discriminating among competing diagnoses is to make system measurements. One way [7, 28] is to insert probes into a circuit, or perform laboratory tests on a patient in order to obtain a sequence of internal measurements, therefore making additional observations. Both Reiter and de Kleer and Williams have proved a variety of results about the measurement. However, they do not make a systematic and formal analysis for their theory of measurement. How can we compute the new diagnosis resulting from a new measurement? How can we compute the new diagnosis based on the information about the old one? The focus of this paper is on providing a basis for the theory of measurement in diagnosis from first principles. Our theory primarily builds upon, and generalizes, the work of Reiter [28] and de Kleer and Williams [7].

In this paper, we provide an efficient incremental method for computing new diagnoses given a new measurement, based on the previous diagnoses predicting the opposite. Our method for computing new diagnoses resulting from a new measurement is also based on the concept of a conflict set. In order to describe our theory, like Reiter, we initially chose first-order logic as a language for representing such a system. But based upon our findings, many different logics will lead to the same theory of measurement in diagnosis presented in this paper. This topic will be addressed in Section 5.3. Whatever one's choice of representation logic is, the description of a system will specify how that system normally behaves on the assumption that all its components are functioning correctly. If we have available an observation of the system's actual behaviour and if this observation conflicts with (namely, is logically inconsistent with) the way the system is meant to behave, then there is a diagnostic problem. The problem is to determine those system components which, when assumed to be functioning abnormally, will explain the discrepancy between the observed and correct system behaviour. Our approach is to compute all possible new diagnoses resulting from a new measurement by using all new minimal conflict sets derived from previous diagnoses. For the sake of completeness, we present a brief review of the pertinent concepts and definitions of Reiter's theory, so that the definitions and theorems from Reiter's paper [28] will be presented essentially unchanged in Section 2.

After proving some preliminary results, including the concept of conflict set

resulting from a measurement, equivalence classes and homogeneous diagnoses, in Sections 3 and 4, a procedure "DIAGNOSE" is derived in Section 5 for computing all diagnoses resulting from a new measurement. This procedure has a number of virtues, not the least of which is its relative independence of the particular logic representing the system being diagnosed. It is incremental, exploiting the iterative nature of diagnosis. And it calculates only all new minimal conflict sets resulting from a new measurement instead of all minimal conflict sets for the faulty system. Furthermore, it calculates only all different new minimal conflict sets resulting from a new measurement.

In Section 7 we consider the relationship of our theory of measurement to other research in this area. Finally, we summarize what we take to be the principal contributions of this work in Section 8.

2. Reiter's consequences

To seek a very general theory of diagnosis, Reiter defines the domain-independent concept of a system which is designed to formalize as abstractly as possible the concept of a component, and the concept of a collection of interacting components.

Definition 2.1. A *system* to be diagnosed is a triple (SD,COMPONENTS,OBS) where:

(1) SD, the system description, is a set of first-order sentences;
(2) COMPONENTS, the system components, is a finite set of constants;
(3) OBS, the system observations, is a finite set of first-order sentences.

Thus, suppose we have determined that a system (SD,COMPONENTS,OBS) is faulty, by which we mean informally that we have made an observation OBS which conflicts with what the system description predicts should happen were all its components behaving correctly. Hence the fact that the observation OBS conflicts with what the system should do were all its components behaving correctly can be formalized by:

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(C_1), \ldots, \neg\, \text{AB}(C_n)\} \tag{2.1}$$

is inconsistent, where $\text{COMPONENTS} = \{C_1, \ldots, C_n\}$, $\text{AB}(\cdot)$ is a distinguished unary predicate indicating that a component is abnormal.

Intuitively, a diagnosis is a conjecture that certain of the components are faulty (ABnormal) and the rest normal, so as to restore consistency to (2.1).

Definition 2.2. A *diagnosis* for (SD,COMPONENTS,OBS) is a minimal set $\Delta \subseteq$ COMPONENTS such that

$$\text{SD} \cup \text{OBS} \cup \{\text{AB}(c) \mid c \in \Delta\} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\}$$

is consistent. A diagnosis is designated by $[\cdot]$.

The following two results are consequences of the definition of a diagnosis.

Theorem 2.3 [28, Proposition 3.3]. *If Δ is a diagnosis for* (SD,COMPONENTS, OBS), *then for each $c_i \in \Delta$,*

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\} \models \text{AB}(c_i)\,.$$

Theorem 2.4 [28, Proposition 3.4]. $\Delta \subseteq$ COMPONENTS *is a diagnosis for* (SD,COMPONENTS,OBS) *iff Δ is a minimal set such that*

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\}$$

is consistent.

In other words, COMPONENTS $- \Delta$ is a diagnosis for (SD,COMPONENTS,OBS) iff Δ is a maximal set such that

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in \Delta\}$$

is consistent.

To achieve the efficiency of computing diagnoses, Reiter proposes a characterization of diagnoses in terms of the concept of a conflict set.

Definition 2.5. A *conflict set* for (SD,COMPONENTS,OBS) is a set $\{c_1, \ldots, c_k\} \subseteq$ COMPONENTS such that

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c_1), \ldots, \neg\, \text{AB}(c_k)\}$$

is inconsistent. A conflict set is designated by $\langle \cdot \rangle$.

A conflict set for (SD,COMPONENTS,OBS) is *minimal* iff no proper subset of it is a conflict set for (SD,COMPONENTS,OBS).

Definition 2.6. Suppose C is a collection of sets. A *hitting set* for C is a set $H \subseteq \bigcup_{S \in C} S$ such that $H \cap S \neq \{\,\}$ for each $S \in C$. A hitting set for C is *minimal* iff no proper subset of it is a hitting set for C.

The following theorem is Reiter's principal characterization of diagnoses, and will provide the basis for computing diagnoses.

Theorem 2.7 [28, Corollary 4.5]. $\Delta \subseteq$ COMPONENTS *is a diagnosis for* (SD,COMPONENTS,OBS) *iff Δ is a minimal hitting set for the collection of minimal conflict sets for* (SD,COMPONENTS,OBS).

Example 2.8. Figure 1 illustrates a device first introduced by Davis [3], and subsequently extensively analyzed by de Kleer and Williams [7]. The device has 5 components: 3 multipliers M_1, M_2, and M_3, and 2 adders A_1 and A_2. The

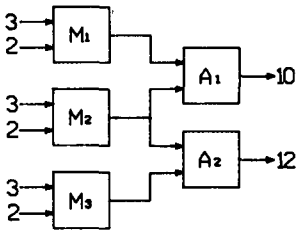


Fig. 1. A familiar circuit.

observation is given by:

$$\text{in1}(M_1) = 3\,, \quad \text{in2}(M_1) = 2\,, \quad \text{in1}(M_2) = 3\,, \quad \text{in2}(M_2) = 2\,,$$
$$\text{in1}(M_3) = 3\,, \quad \text{in2}(M_3) = 2\,, \quad \text{out}(A_1) = 10\,, \quad \text{out}(A_2) = 12\,.$$

We omit a full specification of the system description; it will involve axioms specifying how the components normally behave, axioms describing how the components are interconnected, together with axioms about addition and multiplication of integers, etc.

Since the observation $\text{out}(A_1) = 10$ conflicts with the predicted value $\text{out}(A_1) = 12$, the device is faulty. All minimal conflict sets for this device are $\langle M_1, M_2, A_1 \rangle$ and $\langle M_1, M_3, A_1, A_2 \rangle$, and four diagnoses for this device are $[M_1]$, $[A_1]$, $[M_2, M_3]$ and $[M_2, A_2]$.

Suppose that (SD,COMPONENTS,OBS) has more than one diagnosis. As indicated above, there are four diagnoses. Without further information about the system, one cannot conjecture a unique diagnosis. One way to obtain further information about a system is to perform measurements of some kind e.g. insert a probe into a circuit, or perform a laboratory test on a patient. A measurement MEAS will be defined to be an additional observation (and therefore a finite set of first-order sentences). A system (SD,COMPONENTS,OBS) with MEAS will be written (SD,COMPONENTS,OBS $\cup$ MEAS).

Definition 2.9. A diagnosis Δ for (SD,COMPONENTS,OBS) *predicts* Π (a first-order sentence) iff

$$\text{SD} \cup \text{OBS} \cup \{\text{AB}(c) \mid c \in \Delta\} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\} \models \Pi\,,$$

that is, on the assumption that the components of Δ are all faulty, and the remaining components are all functioning normally, system behaviour Π must hold.

Theorem 2.10 [28, Proposition 5.3]. *A diagnosis* Δ *for* (SD,COMPONENTS,OBS) *predicts* Π *iff*

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\} \models \Pi\,.$$

Theorem 2.11 [28, Theorem 5.7]. *Suppose every diagnosis for* (SD,COMPONENTS,OBS) *predicts one of* Π, $\neg\Pi$. *Then*:

(1) *Every diagnosis for* (SD,COMPONENTS,OBS) *which predicts* Π *is a diagnosis for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

(2) *No diagnosis for* (SD,COMPONENTS,OBS) *which predicts* $\neg\Pi$ *is a diagnosis for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

(3) *Any diagnosis for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *which is not a diagnosis for* (SD,COMPONENTS,OBS) *is a strict superset of some diagnosis for* (SD,COMPONENTS,OBS) *which predicts* $\neg\Pi$. *In other words, any new diagnosis resulting from the new measurement* Π *will be a strict superset of some old diagnosis which predicted* $\neg\Pi$.

Example 2.12. Consider the device of Fig. 1, with the indicated inputs and outputs. Recall that this had four diagnoses: $[M_1]$, $[A_1]$, $[M_2, M_3]$ and $[M_2, A_2]$.

$$[M_1] \text{ predicts } \text{out}(M_2) = 6\,, \qquad [A_1] \text{ predicts } \text{out}(M_2) = 6\,,$$

$$[M_2, M_3] \text{ predicts } \text{out}(M_2) = 4\,, \qquad [M_2, A_2] \text{ predicts } \text{out}(M_2) = 4\,.$$

Suppose that we now measure $\text{out}(M_2)$ and obtain $\text{out}(M_2) = 5$. All four diagnoses predict $\text{out}(M_2) \neq 5$, so that the four old diagnoses are replaced by four new ones (their strict superset): $[M_1, M_2, M_3]$, $[M_1, M_2, A_2]$, $[M_2, M_3, A_1]$ and $[M_2, A_1, A_2]$. On the other hand, if we obtain $\text{out}(M_2) = 6$, then $[M_1]$ and $[A_1]$ remain and become the new diagnoses for (SD,COMPONENTS,OBS $\cup \{\text{out}(M_2) = 6\}$), but $[M_2, M_3]$ and $[M_2, A_2]$ are denied. In some cases, they will be replaced by their strict supersets if these supersets exist.

3. Diagnosis resulting from measurement

The remainder of this paper presents a theory of measurement in diagnosis from first principles, resulting in an efficient diagnostic procedure.

As discussed above, when (SD,COMPONENTS,OBS) has more than one diagnosis, one cannot discriminate among competing diagnoses and conjecture a unique diagnosis without performing measurements. One way is to insert a probe into a circuit or perform a laboratory test on a patient. Having obtained a measurement MEAS, how can we compute diagnoses for (SD,COMPONENTS, OBS $\cup$ MEAS)? A direct natural mechanism for such a computation is to call Reiter's algorithm for (SD,COMPONENTS,OBS $\cup$ MEAS). The obvious problem with this approach is that it is too inefficient for systems with large numbers of components, and that the computation will be redundant.

The objective in this section is to show how to determine all diagnoses for

(SD,COMPONENTS,OBS ∪ MEAS). Our approach to computing all diagnoses for (SD,COMPONENTS,OBS ∪ MEAS) is based upon all new minimal conflict sets resulting from the measurement MEAS together with the old minimal conflict sets for (SD,COMPONENTS,OBS). It is not difficult to see that either Reiter's method or Greiner's corrective method [11] should compute all minimal conflict sets. The alternative used by their procedures is just trying to search for different minimal conflict sets directly. Otherwise the computation can result in an incomplete diagnostician, since it is possible to lose minimal hitting sets, and therefore, diagnoses. For instance, in Example 2.8 above, if we search for conflict sets $\langle M_1, M_3, A_1, A_2\rangle$ and $\langle M_1, M_2, M_3, A_1\rangle$, instead of $\langle M_1, M_2, A_1\rangle$, we will determine the diagnostic candidates: $[M_1]$, $[A_1]$, $[M_3]$, $[M_2, A_2]$. Obviously, the set $[M_3]$ which is not a diagnosis is produced, but the intuitively plausible diagnosis $[M_2, M_3]$ is lost.

Therefore, in our approach, we must study how we determine all diagnoses for (SD,COMPONENTS,OBS ∪ MEAS) in a reasonable way from the previous diagnoses.

Theorem 3.1. *If* (SD,COMPONENTS,OBS) *has only one minimal conflict set, then the diagnosis is a singleton.*

Proof. Suppose that $P = \langle c_1, \ldots, c_k\rangle \subseteq$ COMPONENTS is the only minimal conflict set for (SD,COMPONENTS,OBS).

(1) By the minimality of P, we must have, for each $c_i \in P$, that

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in P - \{c_i\}\}$$

is consistent. Let $P_1 = P - \{c_i\}$, so that P_1 has a maximal strict superset P_1', $c_i \notin P_1'$, such that

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in P_1'\}$$

is consistent. By Theorem 2.4, COMPONENTS $- P_1'$ is a diagnosis. If $P_1' \neq$ COMPONENTS $- \{c_i\}$, then there exists some $c_j \in$ COMPONENTS, such that $c_j \neq c_i$, $c_j \notin P_1'$. By the maximality of P_1', $P_1' \cup \{c_j\}$ will be a conflict set for (SD,COMPONENTS,OBS). Since $c_i \notin P_1' \cup \{c_j\}$, the minimal conflict set derived from $P_1' \cup \{c_j\}$ is not P, contradicting the fact that (SD,COMPONENTS,OBS) has only one minimal conflict set. Hence $P_1' =$ COMPONENTS $- \{c_i\}$, and then COMPONENTS $- P_1' = \{c_i\}$ is a diagnosis for (SD,COMPONENTS,OBS).

(2) Suppose that (SD,COMPONENTS,OBS) has one diagnosis $\Delta = [c_j]$ where $c_j \notin P$, so that we have

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\}$$

is consistent. Obviously P is a strict subset of COMPONENTS $- \Delta$, contradicting the fact that P is a conflict set for (SD,COMPONENTS,OBS).

(3) For any diagnosis $\Delta = [c_{t_1}, \ldots, c_{t_n}]$, if $\Delta \cap P = \{\}$, by the proof of (2), we know that P is not a conflict set; if $\Delta \cap P \neq \{\}$, by the proof of (1), we know that $\Delta = [c_{t_1}]$, $n = 1$, and $c_{t_1} \in P$.

Therefore all diagnoses for (SD,COMPONENTS,OBS) are singletons with the form $[c \mid c \in P]$. □

Definition 3.2. A *conflict set* for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *resulting from the measurement* Π is a set $\{c_1, \ldots, c_k\} \subseteq$ COMPONENTS, such that

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c_1), \ldots, \neg\, \text{AB}(c_k)\}$$

is consistent, and

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg\, \text{AB}(c_1), \ldots, \neg\, \text{AB}(c_k)\}$$

is inconsistent.

A conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from the measurement Π is *minimal* iff no proper subset of it is a conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π.

By the above definition, we must have

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c_1), \ldots, \neg\, \text{AB}(c_k)\} \models \neg\, \Pi\,.$$

Proposition 3.3. *If* $\Delta \subseteq$ COMPONENTS *is a diagnosis for* (SD,COMPONENTS,OBS) *which predicts* $\neg\, \Pi$, *then* COMPONENTS $- \Delta$ *is a conflict set for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *resulting from* Π.

Proof. Since Δ is a diagnosis for (SD,COMPONENTS,OBS) which predicts $\neg\, \Pi$, by Theorem 2.10, we have

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\} \models \neg\, \Pi.$$

Therefore,

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\}$$

is inconsistent, COMPONENTS $- \Delta$ is a conflict set for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$). Since Δ is a diagnosis for (SD,COMPONENTS,OBS),

$$\text{SD} \cup \text{OBS} \cup \{\neg\, \text{AB}(c) \mid c \in \text{COMPONENTS} - \Delta\}$$

is consistent, so that COMPONENTS $- \Delta$ is a conflict set for (SD,COMPONENTS, OBS $\cup \{\Pi\}$) resulting from Π. □

Notice that every subset of a set which is not a conflict set for (SD,COMPONENTS,OBS) is not a conflict set yet, and every superset of a conflict set for (SD,COMPONENTS,OBS) is a conflict set as well.

Definition 3.4. A *conflict set* for (SD,COMPONENTS,OBS) *derived from a conflict set* C is a subset of C which is still a conflict set for (SD,COMPONENTS,OBS). A conflict set for (SD,COMPONENTS,OBS) derived from a conflict set is *minimal* iff no proper subset of it is a conflict set for (SD,COMPONENTS,OBS).

Example 3.5 (*The familiar circuit with the indicated* OBS). One diagnosis $[M_1]$ for (SD,COMPONENTS,OBS) predicts $\text{out}(M_2) = 6$. Suppose we measure $\text{out}(M_2)$ and obtain $\text{out}(M_2) = 5$, so that $[M_1]$ predicts $\text{out}(M_2) \neq 5$. Then COMPONENTS $- \{M_1\} = \{M_2, M_3, A_1, A_2\}$ is a conflict set for (SD,COMPONENTS, OBS $\cup \{\text{out}(M_2) = 5\}$). The minimal conflict sets derived from COMPONENTS $- \{M_1\}$ are $\langle M_2 \rangle$ and $\langle M_3, A_2 \rangle$.

Having the above definitions and propositions, we will propose a method for computing a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π based on the diagnoses for (SD,COMPONENTS,OBS) predicting $\neg \Pi$.

Theorem 3.6. *Every minimal conflict set for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *resulting from* Π *is one of the minimal conflict sets derived from all conflict sets* COMPONENTS $- \Delta_i$ *for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *which are determined by all diagnoses* Δ_i *for* (SD,COMPONENTS,OBS) *predicting* $\neg \Pi$.

Proof. Suppose that all diagnoses for (SD,COMPONENTS,OBS) predicting $\neg \Pi$ are $\Delta_1, \ldots, \Delta_k$, $\Phi_i = \text{COMPONENTS} - \Delta_i$, all minimal conflict sets for (SD, COMPONENTS,OBS $\cup \{\Pi\}$) derived from $\Phi_1, \ldots, \Phi_k$ are $P_1, \ldots, P_n$.

If there is some minimal conflict set P for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π such that $P \neq P_t$ $(t = 1, \ldots, n)$, then $P \not\subseteq \Phi_i$ $(i = 1, \ldots, k)$, $P_t \not\subseteq P$, so that $P \cap \Delta_i \neq \{\}$ for $i \in \{1, \ldots, k\}$. Let $P = \langle c_1, \ldots, c_k, \ldots \rangle$, $c_i \in P \cap \Delta_i$. Since P is a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π, by Definition 3.2,

$$\text{SD} \cup \text{OBS} \cup \{\neg \text{AB}(c) \mid c \in P\}$$

is consistent, and

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in P\}$$

is inconsistent. Construct a maximal superset P' of P, such that

$$\text{SD} \cup \text{OBS} \cup \{\neg \text{AB}(c) \mid c \in P'\}$$

is consistent, then, COMPONENTS $- P'$ is a diagnosis for (SD,COMPONENTS, OBS). Since P is a conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), so is P'. Then,

$$\text{SD} \cup \text{OBS} \cup \{\neg \text{AB}(c) \mid c \in P'\} \models \neg \Pi .$$

By Theorem 2.10, COMPONENTS $- P'$ predicts $\neg \Pi$. Obviously, $c_i \notin$ COMPONENTS $- P'$ $(i = 1, \ldots, k)$, so COMPONENTS $- P' \neq \Delta_i$ $(i = 1, \ldots, k)$, con-

tradicting the fact that all diagnoses for (SD,COMPONENTS,OBS) predicting $\neg \Pi$ are only $\Delta_1, \ldots, \Delta_k$. Then there must exist some Δ_j, such that $\Delta_j \cap P = \{\}$. From this it follows that $P \subseteq \text{COMPONENTS} - \Delta_j$ is one of the minimal conflict sets derived from COMPONENTS $- \Delta_j$, therefore we have $P \in \{P_1, \ldots, P_n\}$. □

Theorem 3.7. *Any minimal conflict set for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *is either a minimal conflict set for* (SD,COMPONENTS,OBS) *or a minimal conflict set for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *resulting from* Π.

Proof. Suppose that P is any one minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). We have

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in P\}$$

is inconsistent.

(1) If P is not a conflict set for (SD,COMPONENTS,OBS), by Definition 3.2, P is a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π.

(2) If P is a conflict set for (SD,COMPONENTS,OBS), then P must be a minimal conflict set for (SD,COMPONENTS,OBS). For if not, P has a strict subset P' such that SD $\cup$ OBS $\cup \{\neg \text{AB}(c) \mid c \in P'\}$ is inconsistent, and so is SD $\cup$ OBS $\cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in P'\}$, contradicting the fact that P is a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). □

Corollary 3.8. $\Delta \subseteq$ COMPONENTS *is a diagnosis for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *iff* Δ *is a minimal hitting set for the collection of minimal conflict sets for* (SD,COMPONENTS,OBS) *together with minimal conflict sets for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *resulting from* Π.

Example 3.9 (*The above circuit*). The minimal conflict sets for (SD,COMPONENTS,OBS) are $\langle M_1, M_2, A_1 \rangle$ and $\langle M_1, M_3, A_1, A_2 \rangle$. The four diagnoses are $\Delta_1 = [M_1]$, $\Delta_2 = [A_1]$, $\Delta_3 = [M_2, M_3]$ and $\Delta_4 = [M_2, A_2]$. Suppose we obtain a measurement $\text{out}(M_2) = 5$. As indicated in Example 2.12, all the four diagnoses predict $\text{out}(M_2) \neq 5$. The minimal conflict sets derived from COMPONENTS $- \Delta_1 = \{M_2, M_3, A_1, A_2\}$ resulting from Π are $\langle M_2 \rangle$, $\langle M_3, A_2 \rangle$. The minimal conflict sets derived from COMPONENTS $- \Delta_2 = \{M_1, M_2, M_3, A_2\}$ resulting from Π are $\langle M_2 \rangle$ and $\langle M_3, A_2 \rangle$. The one derived from COMPONENTS $- \Delta_3 = \{M_1, A_1, A_2\}$ is $\langle M_1, A_1 \rangle$, and so is the one derived from COMPONENTS $- \Delta_4 = \{M_1, M_3, A_1\}$. Computing the minimal hitting sets for the collection of $\langle M_1, M_2, A_1 \rangle$, $\{M_1, M_3, A_1, A_2\rangle$, $\langle M_2 \rangle$, $\langle M_3, A_2 \rangle$ and $\langle M_1, A_1 \rangle$, we obtain that all diagnoses for (SD,COMPONENTS,OBS $\cup \{\text{out}(M_2) = 5\}$) are $[M_1, M_2, M_3]$, $[M_1, M_2, A_2]$, $[M_2, M_3, A_1]$ and $[M_2, A_1, A_2]$.

As we see above, all minimal conflict sets $\langle M_2 \rangle$, $\langle M_3, A_2 \rangle$, $\langle M_1, A_1 \rangle$ resulting from Π are a subset of the old minimal conflict sets $\langle M_1, M_2, A_1 \rangle$, $\langle M_1, M_3, A_1, A_2 \rangle$ for (SD,COMPONENTS,OBS). Therefore, can we compute diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) based only on all minimal conflict sets resulting from Π? The answer is "no". We have the following theorem which reveals the close connection between minimal conflict set for (SD,COMPONENTS,OBS) and minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π.

Theorem 3.10. *For every minimal conflict set for* (SD,COMPONENTS,OBS), *either it is a superset of some minimal conflict set for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *resulting from* Π *or it is not a superset of any minimal conflict set for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *resulting from* Π.

Proof. Suppose that $P = \langle c_1, \ldots, c_k \rangle$ is a minimal conflict set for (SD,COMPONENTS,OBS), then

$$\text{SD} \cup \text{OBS} \cup \{\neg \text{AB}(c_1), \ldots, \neg \text{AB}(c_k)\}$$

is inconsistent. By the minimality of P, for each $c_i \in P$, let $P_i = P - \{c_i\}$, so that we have that

$$\text{SD} \cup \text{OBS} \cup \{\neg \text{AB}(c) \mid c \in P_i\}$$

is consistent.

(1) If there exists some $i \in \{1, \ldots, k\}$, such that $\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in P_i\}$ is inconsistent, then, by Definition 3.2, P_i is a conflict set resulting from Π, so that the minimal conflict sets derived from P_i resulting from Π must be strict subsets of P.

(2) If for every $i \in \{1, \ldots, k\}$, $\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in P_i\}$ is consistent, then P are not a superset of any minimal conflict set resulting from Π. For if not, suppose that Q is a minimal conflict set resulting from Π and is a strict subset of P. Thus

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in Q\}$$

is inconsistent. Since $Q \subset P$, there must exist some $i \in \{1, \ldots, k\}$, $P_i = P - \{c_i\}$, such that $Q \subseteq P_i$. Then

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in P_i\}$$

is inconsistent, a contradiction. Hence, P is not a superset of any minimal conflict set resulting from Π. □

Example 3.11 (*The familiar circuit*). The minimal conflict sets for (SD,COMPONENTS,OBS) are $\langle M_1, M_2, A_1 \rangle$ and $\langle M_1, M_3, A_1, A_2 \rangle$. All four diagnoses

are $\Delta_1 = [M_1]$, $\Delta_2 = [A_1]$, $\Delta_3 = [M_2, M_3]$, $\Delta_4 = [M_2, A_2]$. Suppose that we obtain a measurement out$(M_3) = 8$. Then the three diagnoses Δ_1, Δ_2 and Δ_4 predict out$(M_3) \neq 8$, the diagnosis Δ_3 predicts out$(M_3) = 8$. The minimal conflict set derived from COMPONENTS $- \Delta_4 = \{M_1, M_3, A_1\}$ resulting from Π is $\langle M_3 \rangle$. The ones derived from COMPONENTS $- \Delta_1 = \{M_2, M_3, A_1, A_2\}$ are $\langle M_2, A_2 \rangle$ and $\langle M_3 \rangle$, and so are the ones derived from COMPONENTS $- \Delta_2 = \{M_1, M_2, M_3, A_2\}$. Therefore we have two minimal conflict sets resulting from the measurement out$(M_3) = 8$: $\langle M_2, A_2 \rangle$ and $\langle M_3 \rangle$. Obviously, the old minimal conflict set $\langle M_1, M_3, A_1, A_2 \rangle$ is a superset of $\langle M_3 \rangle$, but the other one, $\langle M_1, M_2, A_1 \rangle$, are not a superset of any minimal conflict set $\langle M_2, A_2 \rangle$ or $\langle M_3 \rangle$.

Corollary 3.12. $\Delta \subseteq$ COMPONENTS *is a diagnosis for* (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$) *iff* Δ *is a minimal hitting set for the collection of minimal conflict sets for* (SD,COMPONENTS,OBS) *which are not a superset of any minimal conflict set resulting from* Π, *together with minimal conflict sets resulting from* Π.

The above approach to computing all diagnoses for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$) depends on the minimal conflict sets for (SD,COMPONENTS,OBS). In practice, for diverse reasons, one may abandon old minimal conflict sets after one has calculated all old diagnoses for (SD,COMPONENTS,OBS). Or viewed from the efficiency, one need not cache old minimal conflict sets after one has determined old diagnoses. Therefore, we should produce a substantial technique for computing new diagnoses based upon old diagnoses instead of old minimal conflict sets.

Theorem 3.13. *All diagnoses for* (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$) *can be generated from all old diagnoses for* (SD,COMPONENTS,OBS) *and all minimal conflict sets for* (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$) *resulting from* Π *as follows*:

(1) *Every diagnosis for* (SD,COMPONENTS,OBS) *which intersects with all minimal conflict sets resulting from* Π *is a diagnosis for* (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$).

(2) *Any diagnosis for* (SD,COMPONENTS,OBS) *which does not intersect with some minimal conflict set(s) resulting from* Π *is replaced by one or more superset candidates which are the union set of the old diagnosis itself and the minimal hitting set for the collection of all minimal conflict sets resulting from* Π *which do not intersect with the old diagnosis. Any tentative new candidate which is subsumed or duplicated by another candidate or diagnosis generated in case* (1) *is eliminated, the remainders are the diagnoses for* (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$).

Proof. Suppose that all diagnoses for (SD,COMPONENTS,OBS) are $\Delta_1, \ldots, \Delta_n$, all minimal conflict sets for (SD,COMPONENTS,OBS) are $Q_1, \ldots, Q_s$, all mini-

mal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π are $P_1, \ldots, P_m$.

Proof of (1). Because Δ_i is a diagnosis for (SD,COMPONENTS,OBS), Δ_i has some overlap with $Q_1, \ldots, Q_s$. Since, for each P_j, $\Delta_i \cap P_j \neq \{\}$, it follows that Δ_i has some overlap with $P_1, \ldots, P_m$. By Theorem 3.7, Δ_i is a hitting set for the collection of all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). It remains only to show that Δ_i is a minimal hitting set. If Δ_i is a singleton, then the subset of Δ_i is empty. Therefore its subset has no overlap with any minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). Suppose that Δ_i is a multiple fault diagnosis, remove any element c from Δ_i. Because Δ_i is an old diagnosis, there must exist some Q_j, such that $c \in Q_j$ and $(\Delta_i - \{c\}) \cap Q_j = \{\}$.

(a) If Q_j is a superset of some P_t, then $(\Delta_i - \{c\}) \cap P_t = \{\}$. Therefore $\Delta_i - \{c\}$ has no overlap with P_t which is a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

(b) If Q_j is not a superset of any P_t, $t = 1, \ldots, m$, then, by Theorem 3.7 and Corollary 3.12, Q_j is a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), and $\Delta_i - \{c\}$ has no overlap with Q_j.

Therefore Δ_i is a minimal hitting set, and thus Δ_i is a diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

Proof of (2). Let Δ have the form $\Delta_i \cup \Delta_{i_1}$, Δ_{i_1} being the minimal hitting set for the collection of all P_t such that $\Delta_i \cap P_t = \{\}$. We write $\Delta_{i_1} = \bigcup_j \{c \mid \forall c \in P_j, P_j \cap \Delta_i = \{\}$, only one $c\}$. Obviously, Δ is a hitting set for the collection of $Q_1, \ldots, Q_s, P_1, \ldots, P_m$. Therefore Δ is a hitting set for the collection of all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). We can notice that the diagnosis generated in case (1) may be a subset of Δ and is an old diagnosis which predicts Π. The candidate Δ generated in case (2) based upon different Δ_j may be the same one. After we have eliminated the tentative candidate which is subsumed or duplicated by another candidate or diagnosis, it remains only to show that the remainder Δ is a minimal hitting set.

For any remainder $\Delta = \Delta_i \cup \Delta_{i_1} = \Delta_i \cup (\bigcup_j \{c \mid \forall c \in P_j, P_j \cap \Delta_i = \{\}$, only one $c\})$, we have Δ_i is a diagnosis for (SD,COMPONENTS,OBS). Remove any element c from Δ.

Case A. If $c \in \Delta_{i_1}$, then $c \notin \Delta_i$. The P_j, which provides the c and for which $P_j \cap \Delta_i = \{\}$, is not hit by $\Delta - \{c\}$. But P_j is a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

Case B. If $c \in \Delta_i$, then $c \notin \Delta_{i_1}$, $\Delta_i \not\subseteq \Delta - \{c\}$.

Case B.1. If Δ has no other one which is generated based on different Δ_j before elimination, then $\Delta - \{c\}$ is not a superset of any diagnosis for (SD,COMPONENTS,OBS). For if not, then suppose that $\Delta - \{c\}$ is a superset of one old diagnosis Δ_j. Thus $\Delta_j \neq \Delta_i$, and there is some element $c' \in \Delta_j$, such that $c' \in \Delta_{i_1}$. Suppose the common part of Δ_i and Δ_j is Δ_{ji}, the common part of Δ_{i_1} and Δ_j is Δ_{ji_1}. Then we have, $\Delta_i = \Delta_{ji} \cup \Delta'_{ji}$, $\Delta_{i_1} = \Delta_{ji_1} \cup \Delta'_{ji_1}$, $\Delta_j = \Delta_{ji} \cup \Delta_{ji_1}$,

$\Delta_j \cap \Delta'_{j_1} = \{\}$, $\Delta_j \cap \Delta'_{ji_1} = \{\}$. Therefore, $\Delta = \Delta_i \cup \Delta_{i_1} = (\Delta_{ji} \cup \Delta'_{ji}) \cup (\Delta_{ji_1} \cup \Delta'_{ji_1}) = (\Delta_{ji} \cup \Delta_{ji_1}) \cup \Delta'_{ji} \cup \Delta'_{ji_1} = \Delta_j \cup \Delta'_{ji} \cup \Delta'_{ji_1}$.

(a) By construction, $\Delta = \Delta_i \cup \Delta_{i_1} = \Delta_i \cup \Delta_{ji_1} \cup \Delta'_{ji_1}$. $\forall c \in \Delta'_{ji_1}$, we have some P which provides the c, such that $P \cap \Delta_{ji} = \{\}$, $P \cap \Delta_{ji_1} = \{\}$. For if $P \cap \Delta_{ji_1} \neq \{\}$, then there is some $c' \in P \cap \Delta_{ji_1}$. Since $P \cap \Delta_i = P \cap (\Delta_{ji} \cup \Delta'_{ji}) = \{\}$, the minimal conflict set which provides c is P while the minimal conflict set which provides c' is P', $c \in \Delta'_{ji_1}$, $c' \in \Delta_{ji_1}$, and Δ is a hitting set for the collection of all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), we have $\Delta' = \Delta_i \cup \Delta_{ji_1} \cup (\Delta'_{ji_1} - \{c\})$ is a hitting set for the collection of all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), and is generated in case (2). Therefore Δ will be eliminated, contradicting the fact that Δ is a remainder.

(b) $\forall c \in \Delta'_{ji}$, we have some P which provides the c, such that $P \cap \Delta_j = \{\}$. For if not, $\Delta' = \Delta_j \cup (\Delta'_{ji} - \{c\}) \cup \Delta'_{ji_1}$ has the same hitting as Δ, and is generated in case (2). Therefore Δ will be eliminated, a contradiction.

(c) By the claims (a) and (b) we have: $\forall c \in \Delta'_{ji}$, there is some P which provides the c, such that $P \cap \Delta_j = \{\}$; $\forall c \in \Delta'_{ji_1}$, there is some P which provides the c, such that $P \cap \Delta_{ji} = \{\}$. Hence, $\Delta = \Delta_j \cup \Delta'_{ji} \cup \Delta'_{ji_1} = \Delta_j \cup \Delta_{j_1}$ can be generated in case (2) based on Δ_j, where $\Delta_{j_1} = \Delta'_{ji} \cup \Delta'_{ji_1}$, contradicting the fact that Δ has no other one which is generated based on different Δ_j before elimination. Therefore, we have that $\Delta - \{c\}$ is not a superset of any old diagnosis. Hence $\Delta - \{c\}$ is not a hitting set for the collection of all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). For if not, $\Delta - \{c\}$ must be a superset of some diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). Therefore, $\Delta - \{c\}$ must be a superset of some old diagnosis, a contradiction.

Case B.2. If Δ has another one which is generated based on different Δ_j before elimination, $\Delta = \Delta_j \cup \Delta_{j_1}$, $c \in \Delta_i$ but $c \notin \Delta_j$. Then by the construction of $\Delta = \Delta_j \cup \Delta_{ji}$, we have $c \in \Delta_{j_1}$. Since there is some P which provides the c, such that $P \cap \Delta_j = \{\}$. Then we have that $(\Delta - \{c\}) \cap P = \{\}$, $\Delta - \{c\}$ is not a hitting set for the collection of all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

Case B.3. If for every other Δ which is generated based on different Δ_j before elimination, $\Delta = \Delta_j \cup \Delta_{j_1}$, such that $c \in \Delta_i \cap \Delta_j$. Then we have $\Delta - \{c\}$ is not a superset of any diagnosis for (SD,COMPONENTS,OBS). For if not, from Case B.1 above, Δ must have another $\Delta = \Delta_k \cup \Delta_{k_1}$, such that $c \in \Delta_i$, but $c \notin \Delta_k$, a contradiction.

Therefore, Δ is a minimal hitting set for the collection of all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). Hence Δ is a diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

Since, for every diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), it is either an old diagnosis for (SD,COMPONENTS,OBS) which predicts Π, or a strict superset

of some old diagnosis which predicts $\neg \Pi$. The former can be generated in case (1), and the latter can be generated in case (2). Hence, all diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) can be generated in cases (1) and (2). □

Example 3.14. In Example 3.9, (SD,COMPONENTS,OBS) has four diagnoses: $\Delta_1 = [M_1]$, $\Delta_2 = [A_1]$, $\Delta_3 = [M_2, M_3]$, $\Delta_4 = [M_2, A_2]$. We obtain a measurement $\Pi = \{\text{out}(M_2) = 5\}$. All minimal conflict sets resulting from Π are: $P_1 = \langle M_2 \rangle$, $P_2 = \langle M_1, A_1 \rangle$, $P_3 = \langle M_3, A_2 \rangle$. Thus, $\Delta_1 \cap P_1 = \{\}$ and $\Delta_1 \cap P_3 = \{\}$. Then the tentative new candidates based on Δ_1 are: $[M_1, M_2, M_3]$, $[M_1, M_2, A_2]$. In the same way, we obtain other tentative new candidates: $[A_1, M_2, M_3]$, $[A_1, M_2, A_2]$ based on Δ_2, $[M_2, M_3, M_1]$, $[M_2, M_3, A_1]$ based on Δ_3, and $[M_2, A_2, M_1]$, $[M_2, A_2, A_1]$ based on Δ_4. After elimination, the remainders which are the all diagnoses for (SD,COMPONENTS,OBS $\cup \{\text{out}(M_2) = 5\}$) are $[M_1, M_2, M_3]$, $[M_1, M_2, A_2]$, $[M_2, M_3, A_1]$, $[M_2, A_1, A_2]$. These are the same results as obtained in Example 3.9. On the other hand, in Example 3.11, suppose we make a measurement $\Pi = \{\text{out}(M_3) = 8\}$. All the minimal conflict sets resulting from Π are $P_1 = \langle M_2, A_2 \rangle$, $P_2 = \langle M_3 \rangle$. Since $\Delta_3 \cap P_1 \neq \{\}$ and $\Delta_3 \cap P_2 \neq \{\}$, we have, by Theorem 3.13(1), that Δ_3 is a new diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). Since $\Delta_4 \cap P_1 \neq \{\}$ and $\Delta_4 \cap P_2 = \{\}$, we obtain, by Theorem 3.13(2), a tentative candidate $[M_2, M_3, A_2]$ based on Δ_4. Obviously, $[M_2, M_3, A_2]$ is subsumed by a new diagnosis $[M_2, M_3]$ generated in case (1). In the same way, we obtain other tentative candidates: $[M_1, M_2, M_3]$, $[M_1, M_3, A_2]$ based on Δ_1, and $[M_2, M_3, A_1]$, $[M_3, A_1, A_2]$ based on Δ_2. After elimination, the remainders $[M_1, M_3, A_2]$ and $[M_3, A_1, A_2]$ are the other new diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). Therefore we have three new diagnoses: $[M_2, M_3]$, $[M_1, M_3, A_2]$ and $[M_3, A_1, A_2]$. It is clear that the previous diagnosis Δ_3 predicting Π remains, the previous diagnoses Δ_1 and Δ_2 are replaced by their strict supersets $[M_1, M_3, A_2]$ and $[M_3, A_1, A_2]$, and the one Δ_4 is denied.

The above example indicates that our Theorem 3.13 provides a powerful method for computing new diagnoses given a new measurement because the previous diagnoses predicting the measurement remain as well as the previous diagnoses predicting the opposite will be replaced by their strict supersets if their supersets exist.

Thus far we have found out the solution to the problem how to determine diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from a measurement Π based on diagnoses for (SD,COMPONENTS,OBS). Since the diagnostic process may not be performed ultimately, namely the new diagnoses may not be unique, we should make a next measurement. Typically, we should make a sequence of measurements $\{\Pi_1, \Pi_2, \ldots, \Pi_n\}$. Given a sequence of measurements $\{\Pi_1, \ldots, \Pi_n\}$, how can we calculate diagnoses for (SD,COMPONENTS, OBS $\cup \{\Pi_1, \ldots, \Pi_n\}$)? Because our method makes no particular demand on

the order of measurements $\Pi_1, \ldots, \Pi_n$, by Theorems 3.6, 3.7 and 3.13, we can easily prove the following theorem.

Theorem 3.15. *All diagnoses for* (SD,COMPONENTS,OBS $\cup \{\Pi_1, \ldots, \Pi_n\}$) *can be incrementally generated, namely iteratively computing new diagnoses given each measurement* Π_i *based on the previous diagnoses for* (SD,COMPONENTS, OBS $\cup \{\Pi_1, \ldots, \Pi_{i-1}\}$), *for* $1 \leq i \leq n$.

Proof. Suppose that we make a sequence of measurements $\Pi_1, \ldots, \Pi_n$. By induction on the number of measurement:

Base. If $n = 1$, then based upon all the diagnoses for (SD,COMPONENTS, OBS), all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi_1\}$) resulting from Π_1 can be computed by Theorem 3.6, and all diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1\}$) can be computed by Theorem 3.13. Thus, if $n = 1$, we are done.

Induction. Let $n = k + 1$. By inductive assumption, we have already incrementally generated all the diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1, \ldots, \Pi_k\}$). Since by the definition of measurement, $\{\Pi_1, \ldots, \Pi_k\}$ can be viewed as a collection of some additional observations, OBS $\cup \{\Pi_1, \ldots, \Pi_k\}$ can be viewed as a new collection of observations: OBS'. Similar to the base case, we can prove that all diagnoses for (SD,COMPONENTS,OBS' $\cup \{\Pi_{k+1}\}$) can be incrementally generated based upon the previous diagnoses for (SD,COMPONENTS,OBS'). □

Example 3.16. As indicated in Example 3.9, suppose we make a measurement $\Pi_1 = \{\text{out}(M_2) = 5\}$ first. All minimal conflict sets resulting from Π_1 are $P_{11} = \langle M_2 \rangle$, $P_{21} = \langle M_1, A_1 \rangle$, $P_{31} = \langle M_3, A_2 \rangle$. All diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1\}$) are $\Delta_{11} = [M_1, M_2, M_3]$, $\Delta_{21} = [M_1, M_2, A_2]$, $\Delta_{31} = [M_2, M_3, A_1]$, $\Delta_{41} = [M_2, A_1, A_2]$. Δ_{11} predicts $\text{out}(M_3) = 7$, Δ_{21} predicts $\text{out}(M_3) = 6$, Δ_{31} predicts $\text{out}(M_3) = 7$, Δ_{41} predicts $\text{out}(M_3) = 6$. Then suppose we make a next measurement $\Pi_2 = \{\text{out}(M_3) = 6\}$. Δ_{11} and Δ_{31} predict the opposite. The minimal conflict set derived from COMPONENTS $- \Delta_{11} = \{A_1, A_2\}$ is $\langle A_2 \rangle$, so is the one derived from COMPONENTS $- \Delta_{31} = \{M_1, A_2\}$. By Theorem 3.13, Δ_{21} and Δ_{41} predicting Π_2 remain and become the new diagnoses. The tentative candidates based on Δ_{11} and Δ_{31} are $[M_1, M_2, M_3, A_2]$ and $[M_2, M_3, A_1, A_2]$ respectively. After elimination, the new diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1, \Pi_2\}$) are $[M_1, M_2, A_2]$ and $[M_2, A_1, A_2]$. On the other hand, suppose we make a measurement $\Pi_1' = \{\text{out}(M_3) = 6\}$ first. Since $\Delta_1 = [M_1]$ predicts $\text{out}(M_3) = 6$, $\Delta_2 = [A_1]$ and $\Delta_4 = [M_2, A_2]$ predict $\text{out}(M_3) = 6$ as well, and $\Delta_3 = [M_2, M_3]$ predicts $\text{out}(M_3) = 8$, we have that the minimal conflict set derived from COMPONENTS $- \Delta_3$ is $\langle M_1, A_1, A_2 \rangle$. Therefore, the minimal conflict set resulting from Π_1' is $P_{11} = \langle M_1, A_1, A_2 \rangle$. By Theorem 3.13, all diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1'\}$) are $\Delta_{11} = [M_1]$,

$\Delta_{21} = [A_1]$, $\Delta_{31} = [M_2, A_2]$. Δ_{11} predicts out(M_2) = 6, Δ_{21} predicts out(M_2) = 6, Δ_{31} predicts out(M_2) = 4. Then suppose we make a next measurement $\Pi_2' = \{\text{out}(M_2) = 5\}$. All previous diagnoses Δ_{11}, Δ_{21} and Δ_{31} predict the opposite. The minimal conflict sets derived from COMPONENTS $- \Delta_{11} = \{M_2, M_3, A_1, A_2\}$ and COMPONENTS $- \Delta_{21} = \{M_1, M_2, M_3, A_2\}$ respectively are the same ones: $\langle M_2 \rangle$, $\langle A_2 \rangle$. The one derived from COMPONENTS $- \Delta_{31} = \{M_1, M_3, A_1\}$ is $\langle M_1, A_1 \rangle$. Therefore all minimal conflict sets resulting from Π_2' are $P_{12} = \langle M_1, A_1 \rangle$, $P_{22} = \langle M_2 \rangle$, $P_{32} = \langle A_2 \rangle$. By Theorem 3.13, the tentative candidates based on Δ_{11}, Δ_{21} and Δ_{31} are $[M_1, M_2, A_2]$, $[M_2, A_1, A_2]$ and $[M_1, M_2, A_2]$, $[M_2, A_1, A_2]$ respectively. After elimination, the new diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1', \Pi_2'\}$) are $[M_1, M_2, A_2]$ and $[M_2, A_1, A_2]$. These are the same results as obtained in the first half of this example.

Example 3.16 indicates that the diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1, \ldots, \Pi_n\}$) are independent of the order of measurements $\Pi_1, \ldots, \Pi_n$ and can be incrementally generated. This is the conclusion that can be drawn from Theorem 3.15. Therefore, our method for computing new diagnoses given a sequence of measurements based on Theorem 3.15 exploits the iterative nature of diagnosis.

Theorem 3.15 is a generalization of the candidate generation procedure of de Kleer and Williams [7, Section 2.5], and provides a formal justification for their procedure. Although, in their research work, they have studied the approach to computing minimal conflict set(s) for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π, their method is different from ours. They compute minimal conflict sets resulting from Π as follows: suppose that quantity x is measured and a value Π is obtained. In terms of SD $\cup$ OBS $\cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in \text{ENV}\}$, they explore every supporting environment ENV, which is identified by a set of assumptions, from the empty set (namely ENV = { }) towards parents sets along the subset–superset lattice, i.e., they explore all of the supporting environments from the bottom towards the root of the lattice. Before a new environment is explored, all other environments which are a subset of the new environment must be explored first. If the environment is inconsistent, namely, it is the supporting environment of any value distinct from the measurement Π, then it is a minimal conflict set and its supersets are not explored. All of the supporting environments of any value distinct from the measurement Π are the minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π. Instead, our approach to computing minimal conflict sets resulting from Π is to determine all minimal conflict sets derived from COMPONENTS $- \Delta$, in which Δ is an old diagnosis for (SD,COMPONENTS,OBS) predicting $\neg \Pi$. In fact, the minimal conflict sets resulting from Π they explored are the same ones as we explored. However, our method is more significant than theirs. In the next section, we will see that we can exploit some properties of the minimal conflict set resulting from Π so as to lead to an enhanced mechanism.

4. Enhanced conflict recognition

In the above section we proved the correctness of our method for computing diagnoses for (SD,COMPONENTS,OBS ∪ MEAS) based on old diagnoses for (SD,COMPONENTS,OBS) and minimal conflict sets for (SD,COMPONENTS,OBS ∪ MEAS) resulting from a measurement MEAS. In fact, we revealed the relationship between old diagnoses and new ones, and proposed a reasonable method for computing new diagnoses, namely less computation is required in the course of the generation of new diagnoses resulting from a measurement.

In this section we go one step further to discuss that the computation of minimal conflict sets resulting from MEAS can be reduced to a minimum, namely only the different minimal conflict sets will be generated. The corresponding theorems provide a basis for our procedure using a pruning device and result in a powerful diagnostic procedure for the diagnostic problem in a domain of a large number of components.

In Example 3.9 above, we noticed that the minimal conflict sets derived from COMPONENTS $- \Delta_1$ and COMPONENTS $- \Delta_2$ respectively are the same, so are the results from COMPONENTS $- \Delta_3$ and COMPONENTS $- \Delta_4$. This is no accident, as the following results show.

Definition 4.1. For the system (SD,COMPONENTS,OBS), $\forall c_1, c_2 \in$ COMPONENTS, c_1 and c_2 are *equivalent* is defined as: c_1 and c_2 hit the same minimal conflict sets for (SD,COMPONENTS,OBS), namely for every minimal conflict set P for (SD,COMPONENTS,OBS), if $c_1 \in P$ then $c_2 \in P$; and if $c_1 \notin P$ then $c_2 \notin P$. This relation is indicated by $c_1 \backsim c_2$. All equivalent elements construct an equivalence class designated by $[\![c_1, c_2, \ldots]\!]$. The elements c_1 and c_2 which are not equivalent are indicated by $c_1 \not\backsim c_2$.

Definition 4.2. The diagnoses Δ_1, Δ_2 for (SD,COMPONENTS,OBS) are defined to be *homogeneous* if: $\forall c_1 \in \Delta_1$, there must exist some $c_2 \in \Delta_2$, such that $c_1 \backsim c_2$, and, $\forall c_1 \in \Delta_2$, there must exist some $c_2 \in \Delta_1$, such that $c_1 \backsim c_2$. The homogeneous diagnoses are indicated by $\Delta_1 \backsim \Delta_2$.

Example 2.8 (*continued*). The circuit has two minimal conflict sets $\langle M_1, M_2, A_1 \rangle$, $\langle M_1, M_3, A_1, A_2 \rangle$, and four diagnoses $[M_1]$, $[A_1]$, $[M_2, M_3]$, $[M_2, A_2]$. Then we have $M_1 \backsim A_1$, $M_2 \backsim M_2$, $M_3 \backsim A_2$, $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![M_3, A_2]\!]$, $[M_1] \backsim [A_1]$, $[M_2, M_3] \backsim [M_2, A_2]$.

Proposition 4.3. *Suppose that Δ is a diagnosis for* (SD,COMPONENTS,OBS) *which predicts $\neg \Pi$. For every $Q \subseteq$* COMPONENTS $- \Delta$ *which is not a conflict set for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$), *there must be a maximal superset Q' of Q which is still a subset of* COMPONENTS $- \Delta$, *such that* SD ∪ OBS ∪ $\{\Pi\}$ ∪ $\{\neg$ AB$(c) \mid c \in Q'\}$ *is consistent.*

Proof. Suppose that Δ is a diagnosis for (SD,COMPONENTS,OBS) which predicts $\neg\, \Pi$. For every diagnosis Δ_1 for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) which is a strict superset of Δ, we have COMPONENTS $- \Delta_1 \subset$ COMPONENTS $- \Delta$.

Suppose that $P_1, \ldots, P_m$ are all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) derived from COMPONENTS $- \Delta$, $Q \subseteq$ COMPONENTS $- \Delta$ is not a conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). Then $P_i \cap \Delta = \{\}$, $i = 1, \ldots, m$. For every minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), if it does not intersect with Δ, then it must be one of $P_1, \ldots, P_m$. The remaining minimal conflict sets must intersect with Δ, if they exist.

Since (COMPONENTS $- \Delta) \cap \Delta = \{\}$, we construct a subset Δ_2 of COMPONENTS $- \Delta$ as follows: $\forall c \in$ COMPONENTS $- \Delta$, if $P_i \not\subseteq \Delta_2 \cup \{c\}$, then add c into Δ_2, $i = 1, \ldots, m$. Then Δ_2 is a maximal subset of COMPONENTS $- \Delta$ which is not a conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). So Δ_2 has the form (COMPONENTS $- \Delta - \bigcup_i P_i) \cup (\bigcup_i Q_i)$, where $Q_i = P_i - \{c\}$, $\forall c \in P_i$. Obviously, selecting a different c in P_i, we will obtain a different Δ_2. Suppose all of the Δ_2 are $\Delta_{2_1}, \ldots, \Delta_{2_s}$. Then the above COMPONENTS $- \Delta_1$ is one of $\Delta_{2_1}, \ldots, \Delta_{2_s}$, and Q is the subset of one of $\Delta_{2_1}, \ldots, \Delta_{2_s}$. Hence, Q must have a maximal superset which is still a subset of COMPONENTS $- \Delta$, such that it is not a conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). □

Theorem 4.4. *Suppose that Δ_1 and Δ_2 are diagnoses for* (SD,COMPONENTS,OBS) *which predict $\neg\, \Pi$ and are homogeneous. If all minimal conflict sets for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *derived from* COMPONENTS $- \Delta_1$ *do not intersect with Δ_2, then all minimal conflict sets for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *derived from* COMPONENTS $- \Delta_2$ *do not intersect with Δ_1.*

Proof. Since $\Delta_1 \backsim \Delta_2$, we can suppose that $\Delta_1 = [\alpha_1, \ldots, \alpha_k] \cup \Delta'$, $\Delta_2 = [\beta_1, \ldots, \beta_k] \cup \Delta'$, $\alpha_i \backsim \beta_i$, $i = 1, \ldots, k$, $\Delta' = \{\}$ or $\Delta' \neq \{\}$.

Suppose some minimal conflict set Q_i derived from COMPONENTS $- \Delta_2$ intersects with Δ_1, $\alpha_i \in Q_i \cap \Delta_1$. By the minimality of Q_i, $Q_{i_1} = Q_i - \{\alpha_i\}$ is not a conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). By Proposition 4.3, there must exist some maximal superset Q'_{i_1} of Q_{i_1}, $Q'_{i_1} \subset$ COMPONENTS $- \Delta_2$, such that

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg\, \text{AB}(c) \mid c \in Q'_{i_1}\}$$

is consistent, so that we have, COMPONENTS $- Q'_{i_1}$ is a diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), and is a strict superset of Δ_2, $\alpha_i \in$ COMPONENTS $- Q'_{i_1}$. Let COMPONENTS $- Q'_{i_1} = \{\beta_1, \ldots, \beta_k, \alpha_i, \ldots\} \cup \Delta'$, so that for every β_j, we have $\alpha_i \not\backsim \beta_j$, $j = 1, \ldots, k$. Since for (SD,COMPONENTS,OBS) we have $\alpha_1 \backsim \beta_1, \ldots, \alpha_k \backsim \beta_k$, but for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) we have $\alpha_i \not\backsim \beta_j$, $j = 1, \ldots, k$. Therefore, for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), there must be some β_l such that $\beta_l \not\backsim \alpha_j$, $j = 1, \ldots, k$. For if not, there must be $\beta_l \backsim \alpha_t$, $\beta_s \backsim \alpha_t$, such that COMPONENTS $- Q'_{i_1}$ is not a diagnosis, a

contradiction. Obviously, $\forall c \in \Delta'$, we have $\beta_l \not\sim c$. Therefore there must exist some minimal conflict set T for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), such that $\beta_l \in T$, and $\forall \alpha_j \notin T$, $j = 1, \ldots, k$. Also, $\forall c \in \Delta'$, we have $c \notin T$. Obviously T is a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) derived from COMPONENTS $- \Delta_1$, and $T \cap \Delta_2 = \{\beta_l\} \neq \{\}$, contradicting the hypothesis of this theorem.

Therefore, every minimal conflict set derived from COMPONENTS $- \Delta_2$ does not intersect with Δ_1. □

Theorem 4.5. *Suppose that Δ_1 and Δ_2 are diagnoses for* (SD,COMPONENTS,OBS) *which predict $\neg \Pi$ and are homogeneous. If all minimal conflict sets for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$) *derived from* COMPONENTS $- \Delta_1$ *do not intersect with Δ_2, then all minimal conflict sets derived from* COMPONENTS $- \Delta_1$, COMPONENTS $- \Delta_2$ *respectively are the same ones.*

Proof. Suppose that all minimal conflict sets derived from COMPONENTS $- \Delta_1$ are $P_1, \ldots, P_m$, all minimal conflict sets derived from COMPONENTS $- \Delta_2$ are $Q_1, \ldots, Q_n$. If $\{P_1, \ldots, P_m\} \neq \{Q_1, \ldots, Q_n\}$, there must be some P_i or Q_i such that $P_i \notin \{Q_1, \ldots, Q_n\}$ or $Q_i \notin \{P_1, \ldots, P_m\}$.

(1) Suppose that there is such a P_i, then there must be some $\alpha \in$ COMPONENTS, such that $\alpha \in P_i$ and $\alpha \notin$ COMPONENTS $- \Delta_2$. For if not, $\forall c \in P_i$, we have $c \in$ COMPONENTS $- \Delta_2$, hence we will predict that P_i is one of $Q_1, \ldots, Q_n$. Since $\alpha \notin$ COMPONENTS $- \Delta_2$, we have $\alpha \in \Delta_2$. Since $\alpha \in P_i$, we have $P_i \cap \Delta_2 \neq \{\}$, a contradiction.

(2) Suppose that there is such a Q_i, then there must be some $\alpha \in$ COMPONENTS, such that $\alpha \in Q_i$ and $\alpha \notin$ COMPONENTS $- \Delta_1$. For if not, we will predict that Q_i is one of $P_1, \ldots, P_m$. Since $\alpha \notin$ COMPONENTS $- \Delta_1$, we have $\alpha \in \Delta_1$. Since $\alpha \in Q_i$, we have $Q_i \cap \Delta_1 \neq \{\}$. However, by Theorem 4.4 we have, $\forall j$, $j = 1, \ldots, n$, $Q_j \cap \Delta_1 = \{\}$, a contradiction.

Therefore, such P_i or Q_i does not exist; and then, we have $\{P_1, \ldots, P_m\} = \{Q_1, \ldots, Q_n\}$, $m = n$. □

Example 3.9 (*continued*). We notice that $\Delta_1 \backsim \Delta_2$, all minimal conflict sets derived from COMPONENTS $- \Delta_1$ are $\langle M_2 \rangle$, $\langle M_3, A_2 \rangle$ which do not intersect with Δ_2, that is, $\Delta_2 \cap \langle M_2 \rangle = \{\}$, $\Delta_2 \cap \langle M_3, A_2 \rangle = \{\}$. Hence all minimal conflict sets derived from COMPONENTS $- \Delta_2$ are also $\langle M_2 \rangle$, $\langle M_3, A_2 \rangle$. The results derived from COMPONENTS $- \Delta_3$ and COMPONENTS $- \Delta_4$ are the same.

The homogeneous diagnoses and the effect of the homogeneous diagnoses on the minimal conflict sets resulting from a measurement are indicated in Figs. 2–4. In the conflict subset tree (addressed in Section 5.2 below in detail), a conflict set is indicated by $\langle \cdot \rangle$, a minimal conflict set is underlined and a leave

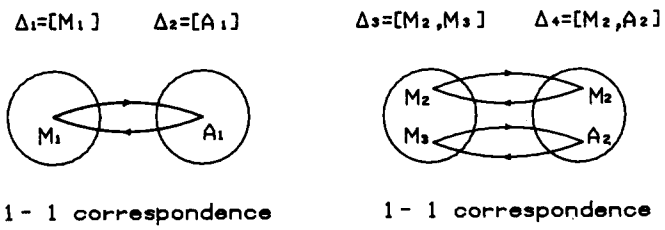


Fig. 2. The homogeneous diagnoses $[M_1] \backsim [A_1]$ and $[M_2, M_3] \backsim [M_2, A_2]$.

node set which is not a conflict set is indicated by $\{\cdot\}$. As we notice, Figs. 3 and 4 show that the two trees have the same structure.

Example 4.6 (*The familiar circuit with the indicated inputs and outputs*). The minimal conflict sets for (SD,COMPONENTS,OBS) are $\langle M_1, M_2, A_1 \rangle$ and $\langle M_1, M_3, A_1, A_2 \rangle$. All diagnoses are $\Delta_1 = [M_1]$, $\Delta_2 = [A_1]$, $\Delta_3 = [M_2, M_3]$, $\Delta_4 = [M_2, A_2]$. Δ_1 predicts out$(M_3) = 6$, Δ_2 predicts out$(M_3) = 6$, Δ_3 predicts out$(M_3) = 8$, Δ_4 predicts out$(M_3) = 6$. Suppose we measure out(M_3) and obtain out$(M_3) = 7$, so that four diagnoses predict out$(M_3) \neq 7$. The minimal conflict sets derived from COMPONENTS $- \Delta_1$ are $\langle M_2, A_2 \rangle$ and $\langle M_3 \rangle$. The ones derived from COMPONENTS $- \Delta_2$ are $\langle M_2, A_2 \rangle$ and $\langle M_3 \rangle$. $\langle M_1, A_1, A_2 \rangle$ is the minimal conflict set derived from COMPONENTS $- \Delta_3$, and $\langle M_3 \rangle$ is the one derived from COMPONENTS $- \Delta_4$. Since $\Delta_1 \backsim \Delta_2$ and $\langle M_2, A_2 \rangle$ and $\langle M_3 \rangle$ derived from COMPONENTS $- \Delta_1$ do not intersect with Δ_2, we have that all minimal conflict sets derived from COMPONENTS $- \Delta_2$ are the same ones. Since $\Delta_3 \backsim \Delta_4$ and $\langle M_1, A_1, A_2 \rangle$ derived from COMPONENTS $- \Delta_3$ intersects with Δ_4, namely $\Delta_4 \cap \langle M_1, A_1, A_2 \rangle = \{A_2\} \neq \{\}$, we have that all minimal conflict sets derived from COMPONENTS $- \Delta_4$ are different from the ones derived from COMPONENTS $- \Delta_3$. There must exist one (e.g. $\langle M_3 \rangle$) that intersects with Δ_3. However, the minimal conflict sets derived from COMPONENTS $- \Delta_3$ which do not intersect with Δ_4, if they exist, must be the same as the ones derived from COMPONENTS $- \Delta_4$ which do not intersect with Δ_3.

The generation of the minimal conflict sets derived from COMPONENTS $- \Delta_3$ and COMPONENTS $- \Delta_4$ is indicated in Figs. 5 and 6 respectively. Figs. 5 and 6 show that the two trees have distinct structures although Δ_3 and Δ_4 (even COMPONENTS $- \Delta_3$ and COMPONENTS $- \Delta_4$) are homogeneous.

Hence, when one computes all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π, one need not consider all COMPONENTS $- \Delta_i$ determined by all diagnoses Δ_i for (SD,COMPONENTS,OBS) which predict $\neg \Pi$. One can generate all minimal conflict sets resulting from Π as follows: first, select any Δ_i, and compute all minimal conflict sets derived from COMPONENTS $- \Delta_i$. Then search for every Δ_j which is some Δ_i's homogeneous

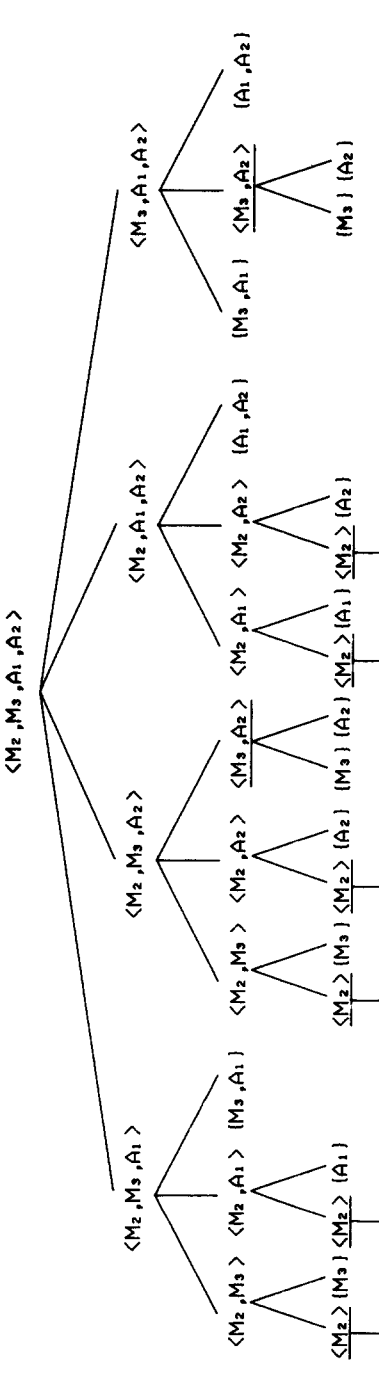


Fig. 3. The minimal conflict sets (underlined) derived from COMPONENTS – $\Delta_1 = \langle M_2, M_3, A_1, A_2 \rangle$.

Fig. 4. The minimal conflict sets (underlined) derived from COMPONENTS – $\Delta_2 = \langle M_1, M_2, M_3, A_2 \rangle$.

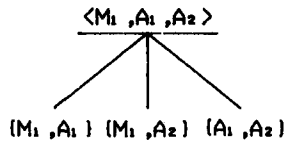


Fig. 5. The minimal conflict set (underlined) derived from COMPONENTS $- \Delta_3 = \langle M_1, A_1, A_2 \rangle$.

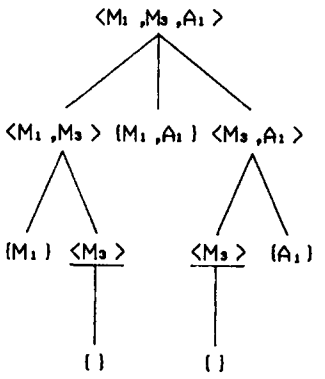


Fig. 6. The minimal conflict set (underlined) derived from COMPONENTS $- \Delta_4 = \langle M_1, M_3, A_1 \rangle$.

diagnosis, determine whether Δ_j intersects with the above minimal conflict sets or not. If not, then COMPONENTS $- \Delta_j$ should not be considered. Otherwise, the minimal conflict sets derived from COMPONENTS $- \Delta_j$ which intersect with Δ_i should be computed and the ones which do not intersect with Δ_i should not be considered. After that, consider the remaining old diagnoses which are not Δ_i's homogeneous sets and the process continues until all minimal conflict sets resulting from Π are generated. Therefore in this approach, one can avoid almost all these inessential computations and improve the efficiency of the algorithm.

If a next measurement Π' should be taken, the above process must be continued for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) so that all diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi, \Pi'\}$) can be generated. Therefore, homogeneous diagnoses and equivalence classes for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) should be computed. As noticed above, in some cases, a minimal conflict set for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) is a remainder of minimal conflict set(s) for (SD,COMPONENTS,OBS). Hence, the equivalence classes determined by only all minimal conflict sets resulting from Π will not be the same ones determined by all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). For instance, in the circuit example above, if we obtain a measurement $\Pi = \{\text{out}(M_3) = 8\}$, all minimal conflict sets resulting from Π are $\langle M_2, A_2 \rangle$, $\langle M_3 \rangle$. The remainder of minimal conflict sets for (SD,COMPONENTS,OBS) is $\langle M_1, M_2, A_1 \rangle$. Obviously, we obtain $M_2 \backsim A_2$ based only on $\langle M_2, A_2 \rangle$ and $\langle M_3 \rangle$. In fact, we have $M_2 \not\backsim A_2$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

If so, one should cache old minimal conflict sets in order to determine the

equivalence classes for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). This is in conflict with our method for computing new diagnoses based on Theorem 3.13. The reason that we propose the method is that we can determine new diagnoses based only on old diagnoses instead of old minimal conflict sets. So, what can we do? Will we compute the new diagnoses based on Corollary 3.12 instead of Theorem 3.13? The answer is "no".

We notice that whether $\alpha \backsim \beta$ or $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS) may be affected by all minimal conflicts for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from Π. However, for every $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), if we determine $\alpha \backsim \beta$ or $\alpha \not\backsim \beta$ based on all minimal conflict sets resulting from Π, we are sure that $\alpha \backsim \beta$ or $\alpha \not\backsim \beta$ is still true for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) respectively. This is because the remainders of minimal conflict sets for (SD,COMPONENTS, OBS), which are also the minimal conflict sets for (SD,COMPONENTS, OBS $\cup \{\Pi\}$), include either both α and β or not any. Suppose that all minimal conflict sets resulting from Π do not include both α and β. If some new diagnosis includes α or β, then there must be a minimal conflict set P for (SD, COMPONENTS,OBS $\cup \{\Pi\}$) which is also the one for (SD,COMPONENTS, OBS), such that $\alpha \in P$, $\beta \in P$, and then $\alpha \backsim \beta$. It follows that $\alpha \backsim \beta$ is still true for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). If no new diagnosis includes α or β, then all minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) do not include both α and β, so that we need not consider the elements α and β for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

Therefore, to cope with the above problem, our further task is to determine the relationship between α and β for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) if we have $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS). If we find an answer to this question, we will move beyond the obstacle indicated above, namely the relationship of elements for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) should be determined based on the old minimal conflict sets (SD,COMPONENTS,OBS).

Theorem 4.7. *For every* $\alpha \not\backsim \beta$ *for* (SD,COMPONENTS,OBS), *we still have* $\alpha \not\backsim \beta$ *for* (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

Proof. Suppose that all minimal conflict sets resulting from Π are $P_1, \ldots, P_m$, and all remainders of minimal conflict sets for (SD,COMPONENTS,OBS) are $Q_1, \ldots, Q_k$. For every $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS):

(1) If there exists some P_i such that $\alpha \in P_i$, $\beta \notin P_i$, or $\alpha \notin P_i$, $\beta \in P_i$, then we can determine $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) based on P_i.

(2) If for every P_i, $\alpha \notin P_i$, $\beta \notin P_i$. Then:

   (a) If there exists some Q_j, such that $\alpha \in Q_j$, $\beta \notin Q_j$, or $\alpha \notin Q_j$, $\beta \in Q_j$, then we can determine $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$) based on Q_j.

(b) If for every Q_j, $\alpha \notin Q_j$, $\beta \notin Q_j$, then every new diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) does not include α and/or β. Hence we need not consider the relationship between α and β for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

(c) If there exists some Q_j such that $\alpha \in Q_j$, $\beta \in Q_j$, the remainders do not include both α and β, if they exist. We must have some new diagnosis for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) which is of the form $\Delta_1 = \Delta_{11} \cup \{\alpha\}$. If $\Delta_2 = \Delta_{11} \cup \{\beta\}$ is not a new diagnosis, we must have $\alpha \not\sim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). On the supposition that $\Delta_2 = \Delta_{11} \cup \{\beta\}$ is a new diagnosis. If Δ_1 is also an old diagnosis, then Δ_{11} is not a superset of any old diagnosis, hence Δ_2 is also an old diagnosis. This results in $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), a contradiction. If Δ_1 is a superset of some old diagnosis, Δ_{11} is not a superset of any old diagnosis. For if not, suppose $\Delta'_{11} \subset \Delta_{11}$ is an old diagnosis, we have $\alpha \notin \Delta'_{11}$. All minimal conflict sets of $Q_1, \ldots, Q_k$ which include α must be hit by some element in Δ'_{11}. Since Δ_{11} intersects with all $P_1, \ldots, P_m$, the subset Δ'_{11} of Δ_{11} intersects with all $Q_1, \ldots, Q_k$, then $\Delta_{11} = \Delta_1 - \{\alpha\}$ intersects with all $P_1, \ldots, P_m$, $Q_1, \ldots, Q_k$, such that Δ_1 is not a new diagnosis, a contradiction. Therefore there must be some old diagnosis Δ_{12}, such that $\alpha \in \Delta_{12}$, $\Delta_{12} \subset \Delta_1$. Let $\Delta_1 = \Delta_{12} \cup \Delta_{22}$, then we have $\Delta_2 = \Delta_{21} \cup \Delta_{22}$, where $\Delta_{21} = (\Delta_{12} - \{\alpha\}) \cup \{\beta\}$. Since Δ_2 is a new diagnosis, $\Delta_1 = \Delta_{12} \cup \Delta_{22}$ is generated in case (2) of Theorem 3.13, we have Δ_{21} as an old diagnosis. This results in $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), a contradiction.

(3) If for every P_i, $\alpha \in P_i$, $\beta \in P_i$. Then we must have that all old diagnoses which include α and/or β must predict Π. For if not, there must exist some P_j, such that $\alpha \notin P_j$ or $\beta \notin P_j$, a contradiction. Hence these old diagnoses are also diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). All new diagnoses which include α and/or β must be old diagnoses, otherwise they will be eliminated in the case (2) of Theorem 3.13.

(a) If there exists some Q_j, such that $\alpha \in Q_j$, $\beta \notin Q_j$, or $\alpha \notin Q_j$, $\beta \in Q_j$, then we can determine $\alpha \not\sim \beta$ for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$) based on Q_j.

(b) If for every Q_j, $\alpha \in Q_j$, $\beta \in Q_j$, then $[\alpha]$ and $[\beta]$ are new diagnoses respectively. Therefore they are old diagnoses, resulting in $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), a contradiction.

(c) If there exists some Q_j such that $\alpha \notin Q_j$, $\beta \notin Q_j$, the remainders include both α and β, if they exist. We must have some new diagnosis which has the form $\Delta_1 = \Delta_{11} \cup \{\alpha\}$, so that Δ_1 is also an old diagnosis. If $\Delta_2 = \{\Delta_1 - \{\alpha\}) \cup \{\beta\}$ is a new diagnosis, then Δ_2 is also an old diagnosis, resulting in $\alpha \backsim \beta$ for (SD,COMPONENTS,

OBS), a contradiction. If Δ_2 is not a new diagnosis, we must have $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

(4) If there exists some P_i such that $\alpha \not\in P_i$, $\beta \not\in P_i$, and the remainders of $P_1, \ldots, P_m$ include both α and β, namely $\alpha \backsim \beta$ will be determined based on the remainders. Then there must be some old diagnosis Δ which includes α and/or β, such that P_i is derived from COMPONENTS $- \Delta$ and Δ predicts $\neg \Pi$. For if not, all old diagnoses which include α and/or β will predict Π. By the proof of (3), we have $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). If $\alpha \in \Delta$ and $\beta \in \Delta$, then we have a new diagnosis which is a strict superset of Δ, so that we can determine $\alpha \not\backsim \beta$ for(SD,COMPONENTS,OBS $\cup \{\Pi\}$). Suppose $\alpha \in \Delta$ and $\beta \not\in \Delta$, we can generate a new diagnosis $\Delta_1 = \Delta \cup (\bigcup_j \{c \mid \forall c \in P_j,\ P_j \cap \Delta = \{\}$, only one $c\})$. If $\Delta_2 = (\Delta_1 - \{\alpha\}) \cup \{\beta\}$ is a new diagnosis, then $\Delta' = (\Delta - \{\alpha\}) \cup \{\beta\}$ is also an old diagnosis, resulting in $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), a contradiction. If Δ_2 is not a new diagnosis, we must have $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). □

Example 4.8 (*The familiar circuit*). All minimal conflict sets for (SD,COMPONENTS,OBS) are $P_{10} = \langle M_1, M_2, A_1 \rangle$, $P_{20} = \langle M_1, M_3, A_1, A_2 \rangle$. All four diagnoses are $\Delta_{10} = [M_1]$, $\Delta_{20} = [A_1]$, $\Delta_{30} = [M_2, M_3]$, $\Delta_{40} = [M_2, A_2]$. Thus we have: (1) equivalent elements $M_1 \backsim A_1$, $M_2 \backsim M_2$, $M_3 \backsim A_2$; (2) equivalence classes $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![M_3, A_2]\!]$; and (3) homogeneous diagnoses $[M_1] \backsim [A_1]$, $[M_2, M_3] \backsim [M_2, A_2]$. As indicated in the first half of the Example 3.16, suppose we make a measurement $\Pi_1 = \{\text{out}(M_2) = 5\}$ first. Then the minimal conflict sets resulting from Π_1 are $P_{11} = \langle M_2 \rangle$, $P_{21} = \langle M_1, A_1 \rangle$, $P_{31} = \langle M_3, A_2 \rangle$. The diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1\}$) are $\Delta_{11} = [M_1, M_2, M_3]$, $\Delta_{21} = [M_1, M_2, A_2]$, $\Delta_{31} = [M_2, M_3, A_1]$, $\Delta_{41} = [M_2, A_1, A_2]$. By Theorem 4.7, we will only consider the effect of P_{11}, P_{21} and P_{31} on the old equivalence classes $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![M_3, A_2]\!]$. In other words, we need not consider the relationship between previous elements (e.g. M_1 and M_2) which were not equivalent. Namely, we need not worry about the generation of new equivalence classes (e.g. $[\![M_1, M_2]\!]$) made up of previous elements which were not equivalent. Thus we have: (1) equivalent elements $M_1 \backsim A_1$, $M_2 \backsim M_2$, $M_3 \backsim A_2$; (2) equivalence classes $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![M_3, A_2]\!]$; and (3) homogeneous diagnoses $\Delta_{11} \backsim \Delta_{21} \backsim \Delta_{31} \backsim \Delta_{41}$. Then suppose we make a next measurement $\Pi_2 = \{\text{out}(M_3) = 6\}$. The minimal conflict sets resulting from Π_2 are $P_{12} = \langle A_2 \rangle$. The diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi_1, \Pi_2\}$) are $\Delta_{12} = [M_1, M_2, A_2]$, $\Delta_{22} = [M_2, A_1, A_2]$. Thus we have: (1) equivalent elements $M_1 \backsim A_1$, $M_2 \backsim M_2$, $M_3 \backsim M_3$, $A_2 \backsim A_2$; (2) equivalence classes $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![M_3]\!]$, $[\![A_2]\!]$; and (3) homogeneous diagnoses $[M_1, M_2, A_2] \backsim [A_1, M_2, A_2]$.

On the other hand, the same results can be drawn based on the minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi_1, \Pi_2\}$). As we notice above, the

minimal conflict sets for (SD,COMPONENTS,OBS) are $P_{10} = \langle M_1, M_2, A_1 \rangle$, $P_{20} = \langle M_1, M_3, A_1, A_2 \rangle$. The minimal conflict sets resulting from Π_1 are $P_{11} = \langle M_2 \rangle$, $P_{21} = \langle M_1, A_1 \rangle$, $P_{31} = \langle M_3, A_2 \rangle$. And the minimal conflict set resulting from Π_2 is $P_{12} = \langle A_2 \rangle$. Therefore, the minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi_1, \Pi_2\}$) are $\langle M_1, M_2, A_1 \rangle$, $\langle M_1, M_3, A_1, A_2 \rangle$, $\langle M_2 \rangle$, $\langle M_1, A_1 \rangle$, $\langle M_3, A_2 \rangle$ and $\langle A_2 \rangle$. After simplification, the remainders are $\langle M_2 \rangle$, $\langle M_1, A_1 \rangle$ and $\langle A_2 \rangle$. Based on these minimal conflict sets (namely $\langle M_2 \rangle$, $\langle M_1, A_1 \rangle$ and $\langle A_2 \rangle$), we can determine: $M_1 \backsim A_1$, $M_2 \backsim M_2$, $A_2 \backsim A_2$, $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![A_2]\!]$.

This example provides a justification for our method for determining the relationship of elements given a sequence of measurements.

As a consequence, when one determines the relationship of elements for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), only does one need to consider the effect of all minimal conflict sets resulting from Π on every $\alpha \backsim \beta$ for (SD,COMPONENTS, OBS), not the effect on every $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS). That is:

(1) For every $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), if one determines $\alpha \backsim \beta$ based on all minimal conflict sets $P_1, \ldots, P_m$ resulting from Π, then one has $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). If one determines $\alpha \not\backsim \beta$ based on $P_1, \ldots, P_m$, then one has $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$).

(2) For some $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), if all $P_1, \ldots, P_m$ do not include both α and β, then one keeps $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$).

(3) For every $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS), one must have $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$), and pay no particular attention to whether $\alpha \backsim \beta$ or $\alpha \not\backsim \beta$ based on $P_1, \ldots, P_m$.

(4) For every new $\alpha \backsim \beta$ determined by all minimal conflict sets $P_1, \ldots, P_m$ resulting from Π, if α and/or β do not occur in the previous equivalence classes, namely α and/or β are not the members of any old minimal conflict set, then one keeps $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

Of course, the following approach is reasonable, that is, one determines the new relationship of elements for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) based only on $P_1, \ldots, P_m$. If, for some $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS), one will determine $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) based on $P_1, \ldots, P_m$, that is not a question, since there must not be new homogeneous diagnoses Δ_1 and Δ_2 such that $\alpha \in \Delta_1$ and $\beta \in \Delta_2$. That is, for any new diagnoses Δ_1 and Δ_2, if $\alpha \in \Delta_1$, $\beta \in \Delta_2$, then we must have $\Delta_1 \not\backsim \Delta_2$.

Thus far, we have provided the solution to the problem indicated above, that is, how to determine the new relationship of elements for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) based on all minimal conflict sets resulting from a measurement Π. Our enhanced mechanism for computing new diagnoses based on

Theorems 3.13 and 4.7 is independent of old minimal conflict sets. Therefore we can calculate new diagnoses based only on old diagnoses.

When a next measurement Π' is obtained, one can continue the above process to determine all diagnoses and equivalence classes for (SD,COMPONENTS,OBS $\cup \{\Pi, \Pi'\}$) based on all diagnoses and equivalence classes for (SD,COMPONENTS,OBS $\cup \{\Pi\}$). In the course of computation, however, only old diagnoses instead of old minimal conflict sets are used. So it is not necessary to cache old minimal conflict sets and one can give up all minimal conflict sets resulting from Π after determining all new diagnoses and new equivalence classes. This allows one to determine the new diagnoses with as few computations as possible.

5. Diagnostic algorithm

Thus far we have described a very general diagnostic strategy for handling multiple faults resulting from measurement. We will develop our procedure for computing all diagnoses for (SD,COMPONENTS,OBS $\cup$ MEAS) in this section. Our approach to computing diagnoses resulting from a measurement MEAS is based upon Theorems 3.6, 3.13, 4.5 and 4.7. Therefore we require equivalence classes, homogeneous diagnoses and minimal conflict sets resulting from MEAS.

5.1. Equivalence class and homogeneous diagnosis

Suppose that we have the collection of all minimal conflict sets, EC = $\{CS_1, \ldots, CS_n\}$. Select any one CS_i, consider one element α in CS_i. Search for all elements β in CS_i equivalent to α, namely α and β hit the same members of EC. If such β exists, we have one equivalence class $[\![\alpha, \beta, \ldots]\!]$. Otherwise, α is equivalent to itself, and we have a single equivalence class $[\![\alpha]\!]$. Then we remove α and its equivalent elements from each of the members of EC, search for a next equivalence class in the remaining members of EC until all equivalence classes have been determined, namely the remaining members of EC are empty sets.

After searching for all equivalence classes, we begin to determine all homogeneous diagnoses. Suppose that we have the collection of all diagnoses, HD = $\{\Delta_1, \ldots, \Delta_m\}$. Select any one Δ_i, search for all Δ_i's homogeneous diagnoses Δ_j in HD, namely search for those members Δ_j in HD such that: for every $\alpha \in \Delta_i$, there must be $\beta \in \Delta_j$, such that $\alpha \backsim \beta$; vice versa, for every $\alpha \in \Delta_j$, there must be $\beta \in \Delta_i$, such that $\alpha \backsim \beta$. If such a Δ_j exists, we have homogeneous diagnoses $\Delta_i \backsim \Delta_j$. Otherwise, Δ_i is homogeneous itself, and we have a single homogeneous diagnosis $\Delta_i \backsim \Delta_i$. Then we remove Δ_i and its homogeneous diagnoses from HD, search for a next homogeneous diagnoses in

the remaining members of HD until all homogeneous diagnoses have been determined, namely the collection HD is empty.

In Example 3.9 above, for instance, all minimal conflict sets for (SD,COMPONENTS,OBS) are $CS_1 = \langle M_1, M_2, A_1 \rangle$, $CS_2 = \langle M_1, M_3, A_1, A_2 \rangle$. All diagnoses for (SD,COMPONENTS,OBS) are $\Delta_1 = [M_1]$, $\Delta_2 = [A_1]$, $\Delta_3 = [M_2, M_3]$, $\Delta_4 = [M_2, A_2]$. Let $EC = \{\langle M_1, M_2, A_1 \rangle, \langle M_1, M_3, A_1, A_2 \rangle\}$. Selecting any element, e.g. M_1, from CS_1, we can determine that its equivalent element is A_1. Therefore, we have one equivalence class $[\![M_1, A_1]\!]$. After removing M_1, A_1 from CS_1 and CS_2, we have the remaining $EC = \{\langle M_2 \rangle, \langle M_3, A_2 \rangle\}$. Search for a next equivalence class from the members of EC. Suppose that we obtain the next equivalence class $[\![M_2]\!]$. Continuing the above process cyclically, we will determine the last equivalence class $[\![M_3, A_2]\!]$ from the remaining member $\langle M_3, A_2 \rangle$ of EC, where $EC = \{\langle\ \rangle, \langle M_3, A_2 \rangle\}$. Hence, we obtain three equivalence classes $[\![M_1, A_1]\!]$, $[\![M_2]\!]$ and $[\![M_3, A_2]\!]$. Since, $M_1 \in \Delta_1$ and $A_1 \in \Delta_2$ are equivalent, Δ_1 and Δ_2 are homogeneous. Since $M_2 \in \Delta_3$ and $M_2 \in \Delta_4$ are equivalent (the same one), as well as $M_3 \in \Delta_3$ and $A_2 \in \Delta_4$ are equivalent, Δ_3 and Δ_4 are homogeneous. Hence we have two homogeneous diagnoses $\Delta_1 \backsim \Delta_2$, and $\Delta_3 \backsim \Delta_4$.

After having determined all equivalence classes and homogeneous diagnoses based on old minimal conflict sets and old diagnoses, we are ready for straightforwardly calculating all different minimal conflict sets for (SD,COMPONENTS,OBS ∪ MEAS) resulting from a measurement MEAS. Generating all minimal conflict sets resulting from MEAS and all diagnoses for (SD,COMPONENTS,OBS ∪ MEAS) based upon Theorems 3.13 and 4.5, we can determine new equivalence classes and homogeneous diagnoses for (SD,COMPONENTS,OBS ∪ MEAS) as follows (discussed in the above section): for every $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS), we keep $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS ∪ MEAS). For every $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS), we will determine the new relationship between α and β based upon all minimal conflict sets resulting from MEAS. If we determine $\alpha \backsim \beta$ based upon all minimal conflict sets resulting from MEAS, then we will have $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS ∪ MEAS). If we determine $\alpha \not\backsim \beta$, then we will keep $\alpha \not\backsim \beta$ for (SD,COMPONENTS,OBS ∪ MEAS). For every new $\alpha \backsim \beta$ determined by all minimal conflict sets resulting from MEAS, namely α and/or β do not occur in the previous equivalence classes, we keep $\alpha \backsim \beta$ for (SD,COMPONENTS,OBS ∪ MEAS).

In Example 3.9 above, suppose that we obtain a measurement $\Pi = \{\text{out}(M_2) = 5\}$. Since we have all minimal conflicts resulting from Π, namely $\langle M_1, A_1 \rangle$, $\langle M_3, A_2 \rangle$, $\langle M_2 \rangle$, we will determine new equivalence classes $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![M_3, A_2]\!]$. In Example 4.6 above, however, we make a measurement $\Pi = \{\text{out}(M_3) = 7\}$ and obtain all minimal conflicts resulting from Π: $\langle M_1, A_1, A_2 \rangle$, $\langle M_2, A_2 \rangle$, and $\langle M_3 \rangle$. Therefore, we will determine new equivalence classes $[\![M_1, A_1]\!]$, $[\![M_2]\!]$, $[\![M_3]\!]$, and $[\![A_2]\!]$. In the course of generation of the new equivalence classes, we only need to consider the effect

of all minimal conflict sets $\langle M_1, A_1, A_2\rangle$, $\langle M_2, A_2\rangle$, $\langle M_3\rangle$ on the old equivalence classes $[\![M_1, A_1]\!]$ and $[\![M_3, A_2]\!]$.

5.2. *Minimal conflict set*

Our approach to computing all minimal conflict sets resulting from a measurement Π is to determine all different minimal conflict sets derived from COMPONENTS $- \Delta_i$ in which Δ_i is a diagnosis for (SD,COMPONENTS,OBS) predicting $\neg \Pi$. We compute these minimal conflict sets by constructing a conflict set tree rooted by COMPONENTS $- \Delta_i$. In the tree, a node is labeled by a subset of the root set COMPONENTS $- \Delta_i$. Obviously, in diagnosis, if a set of components is a conflict, every superset of that set must be a conflict as well. If a set is not a conflict, every subset of that set must not be a conflict yet. Therefore, whenever we determine that a set is not a conflict, we need not consider any subset of that set. This corresponds to stopping the generation of the successors of that node indicated by that set in the conflict set tree. If every subset of a conflict set is not a conflict, then that conflict set is minimal.

Definition 5.1. Suppose F is a conflict set. A CS-tree *rooted by* F, call it T, is a node-labeled tree with the following properties:

(1) Its root is labeled by $\{\}$ if F is empty. Otherwise, its root is labeled by F.
(2) Each node n of T is labeled by a subset S of F. If the set S is not a conflict set, the node n has no successor nodes in T. If the set S is a conflict set, then for each $\sigma \in S$, n has a successor node n_σ such that n_σ is labeled by $S - \{\sigma\}$.
(3) For each node n of T, if its label set S is not a conflict set, then it is indicated by $\{\cdot\}$. If the set S is a conflict set, then it is indicated by $\langle\cdot\rangle$. Further, if the set S is a minimal conflict set, then it is underlined.

Example 5.2. In Example 3.9 above, for instance, COMPONENTS $- \Delta_1 = \langle M_2, M_3, A_1, A_2\rangle$ is a conflict set for (SD,COMPONENTS,OBS $\cup \{\text{out}(M_2) = 5\}$). Fig. 7 is a CS-tree rooted by $\langle M_2, M_3, A_1, A_2\rangle$.

The problem with this approach is that computing all subsets of one conflict set is in general exponential. It is impractical and too inefficient for systems with large numbers of components. To achieve the needed efficiency, our objective is to determine various tree pruning techniques to allow us to generate as small a subtree of a CS-tree as is possible, while preserving the property that the subtree so generated will give us all minimal conflict sets derived from F. In addition, we wish to minimize the number of accesses to F required to generate this subtree, where by an access to F we mean the computation required to determine whether the label set of a node is a conflict set or not by a call to an underlying theorem prover. The role of the theorem

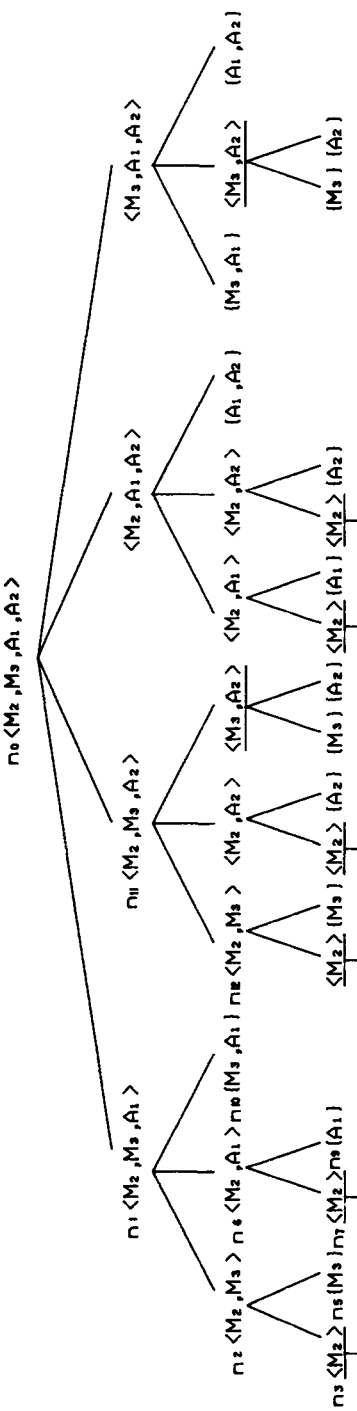


Fig. 7. A CS-tree rooted by $\langle M_2, M_3, A_1, A_2 \rangle$.

prover is simply to determine whether a set is a conflict set or not. Obviously, the empty set { } requires no access to the theorem prover because it cannot be a conflict set. Therefore, a single conflict set must be a minimal conflict set.

The natural way to reduce accesses to F in generating a CS-tree is to reuse node labels which have already been computed. For example, if the CS-tree of Fig. 7 was generated depth-first, then node n_{12} could have been assigned the same label as n_2, namely $\langle M_2, M_3 \rangle$. This means that the subtree rooted at n_2 and n_{12} respectively could be identically generated. Thus, n_{12}'s subtree is redundant, and n_{12} requires no access to F, and we can close node n_{12}.

Another way to reduce accesses to F is to take advantage of the minimal conflict set which has already been computed. If $S \subseteq F$ and $S' \subseteq F$ with S a minimal conflict set and a proper subset of S', then S' has the same minimal conflict set as S. The node labeled by S' requires no access to F, and it can be removed from the tree while preserving the property that the resulting pruned tree will yield all minimal conflict sets derived from F. For example, in Fig. 7, node n_6 is an example of such a node which we can close.

We summarize our method for generating a pruned CS-tree rooted by F as follows:

(1) Generate the CS-tree depth-first, generating all nodes at any fixed level in the tree in root-to-left child-to-right child order, namely descending to generate the successor of one node before generating the adjacent nodes of the node.

(2) Tree pruning:
   (a) If node n has already been labeled by a set $S \subseteq F$, and if node n' is such a new node that it will be labeled by S, then we close node n'. Such a node n' requires no access to F. We indicate such a closed node in the tree by marking it with "*".
   (b) If node n will be labeled by a set $S \subseteq F$, and if S is a proper superset of some minimal conflict set(s) which have already been generated in the tree, then we do not generate any successors of n and close node n. We indicate such a closed node in the tree by marking it with "×".

Figure 8(a) depicts such a pruned CS-tree for the example of Fig. 7, where we consider the subset $\langle M_2, M_3, A_1 \rangle$ of $\langle M_2, M_3, A_1, A_2 \rangle$ first. If it is not a conflict set, then the other subsets $\langle M_2, M_3, A_2 \rangle$, $\langle M_2, A_1, A_2 \rangle$ and $\langle M_3, A_1, A_2 \rangle$ are considered in turn. If it is a conflict set, then the subset $\langle M_2, M_3 \rangle$ of it is considered first. This process continues until we generate the minimal conflict set $\langle M_2 \rangle$. After that, we only consider those sets which are not a superset of $\langle M_2 \rangle$. Namely, $\{M_3\}$ is considered first and then $\{M_3, A_1\}$. We do not generate the subset $\langle M_2, A_1 \rangle$ of $\langle M_2, M_3, A_1 \rangle$. This conflict recognition continues until we find out all minimal conflict sets derived from

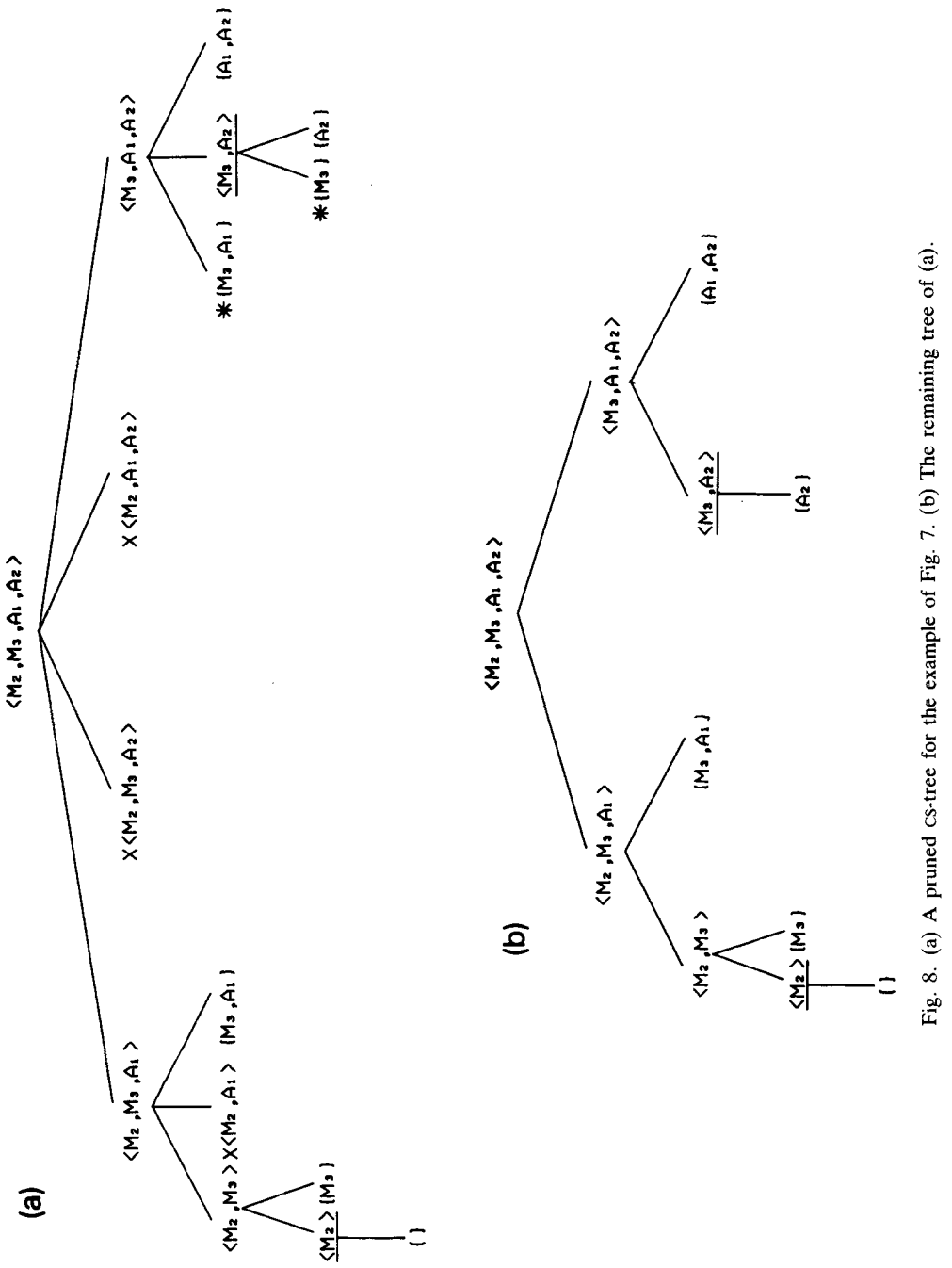


Fig. 8. (a) A pruned CS-tree for the example of Fig. 7. (b) The remaining tree of (a).

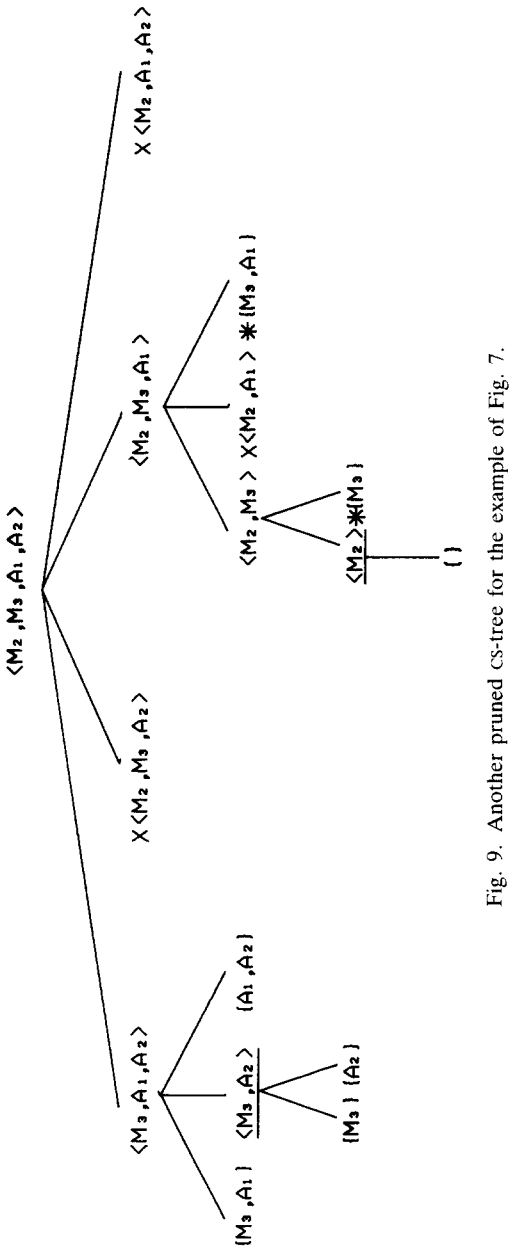


Fig. 9. Another pruned CS-tree for the example of Fig. 7.

$\langle M_2, M_3, A_1, A_2 \rangle$. If we start with the subset $\langle M_3, A_1, A_2 \rangle$ of $\langle M_2, M_3, A_1, A_2 \rangle$ in the construction of the pruned CS-tree, we will construct another pruned CS-tree shown in Fig. 9. If we only use the pruning rule (2a) to construct the tree, we will generate such a pruned CS-tree for the example of Fig. 7 as Fig. 10(a) shows.

Comparisons between Figs. 7, 8(b) and Fig. 10(b) make us obtain:

(1) without the pruning rules (2a) and (2b), the construction of such a CS-tree requires 32 accesses to the theorem prover;
(2) only using the pruning rule (2a), we need 14 accesses to the theorem prover;
(3) with our method for constructing a CS-tree, namely using the two pruning rules (2a) and (2b), we only require 9 accesses to the theorem prover.

As a consequence, our method for generating all minimal conflict sets derived from a conflict set prevents us from the NP-hard problem and, therefore, enhances our mechanism for computing new diagnoses given a new measurement based upon the previous diagnoses predicting the opposite.

In fact, our minimal conflict sets derived from all COMPONENTS $- \Delta_i$, in which Δ_i is a previous diagnosis predicting $\neg \Pi$, are the minimal supporting environments, used in de Kleer's procedure [7], of any value distinct from the measurement Π. It differs only in that de Kleer calculates these minimal supporting environments from the bottom (namely the empty set ENV $= \{\}$) towards the root through a conflict superset lattice which is generated breadth-first in terms of SD $\cup$ OBS $\cup \{\Pi\} \cup \{\neg \text{AB}(c) \mid c \in \text{ENV}\}$.

As indicated above, in the course of construction of a pruned CS-tree, we use the two pruning rules to avoid unnecessary access to the theorem prover. Then, why do we not simply deal with every pruned CS-tree rooted by every COMPONENTS $- \Delta_i$ in the same way? That is, why do we not construct a pruned CS-tree by using node labels already generated in another pruned CS-tree instead of using equivalence classes and homogeneous diagnoses? Of course, that is a nature approach. The reason we did not do this is that, as we have already remarked, for our purpose we will propose a more efficient procedure, since constructing a new pruned CS-tree will require a lot of computation.

Therefore, we will supplement our mechanism by adding a pruning rule (2c) as follows:

(2) Tree pruning (continued):
  (c) If all minimal conflict sets generated in a pruned CS-tree rooted by COMPONENTS $- \Delta$ do not intersect with Δ', where Δ and Δ' are homogeneous diagnoses and both predict the opposite of a given measurement, then the pruned CS-tree rooted by COMPONENTS $- \Delta'$ need not be constructed at all.

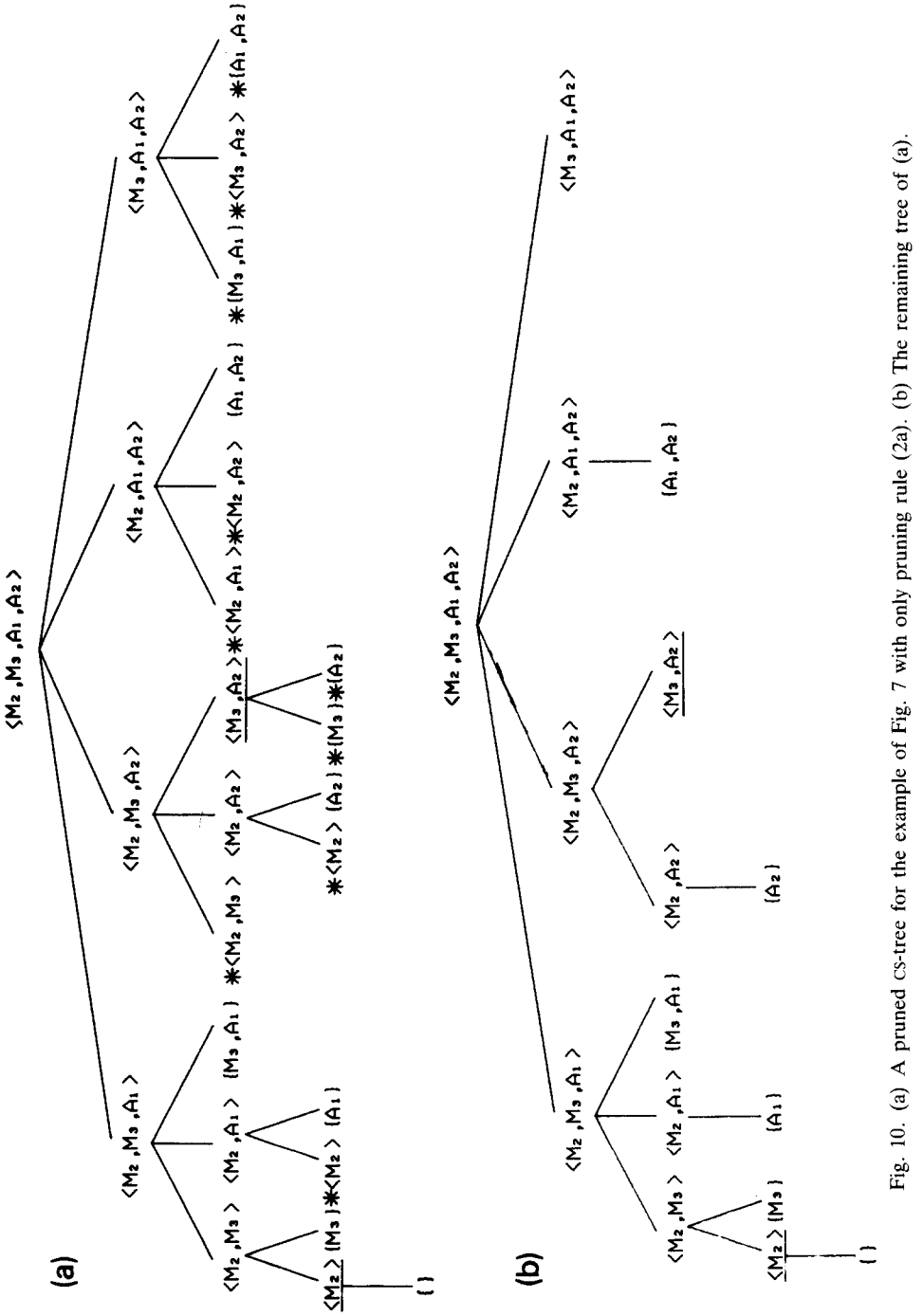


Fig. 10. (a) A pruned CS-tree for the example of Fig. 7 with only pruning rule (2a). (b) The remaining tree of (a).

5.3. Diagnostic candidates

Having computed all old diagnoses for (SD,COMPONENTS,OBS) and all new minimal conflict sets for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) resulting from a measurement Π, we can determine all diagnostic candidates based on Theorem 3.13. After elimination, namely, any tentative new candidate which is subsumed or duplicated by another one is eliminated, the remaining candidates are all diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$).

In Example 3.9, we obtain a measurement $\Pi = \{\text{out}(M_2) = 5\}$. The old diagnoses are $\Delta_1 = [M_1]$, $\Delta_2 = [A_1]$, $\Delta_3 = [M_2, M_3]$ and $\Delta_4 = [M_2, A_2]$. The minimal conflict sets resulting from Π are $P_1 = \langle M_2 \rangle$, $P_2 = \langle M_1, A_1 \rangle$ and $P_3 = \langle M_3, A_2 \rangle$. By Theorem 3.13, the candidates based on Δ_1 are $[M_1, M_2, M_3]$, $[M_1, M_2, A_2]$. The ones based on Δ_2 are $[M_2, M_3, A_1]$, $[M_2, A_1, A_2]$. The ones based on Δ_3 and Δ_4 are $[M_1, M_2, M_3]$, $[M_2, M_3, A_1]$ and $[M_1, M_2, A_2]$, $[M_2, A_1, A_2]$, respectively. After elimination, we obtain all diagnoses for (SD,COMPONENTS, OBS $\cup \{\Pi\}$): $[M_1, M_2, M_3]$, $[M_1, M_2, A_2]$, $[M_2, M_3, A_1]$, and $[M_2, A_1, A_2]$.

Based on all minimal conflict sets resulting from Π, we can determine new equivalence classes and new homogeneous diagnoses for (SD,COMPONENTS, OBS $\cup \{\Pi\}$). When obtaining a next measurement Π', we continue the above process so as to determine all diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi, \Pi'\}$). This process continues until the unique diagnosis is obtained.

Thus, we have proved the correctness of the following procedure.

Procedure DIAGNOSE(SD,COMPONENTS,OBS)
Input: SD is a set of descriptions of a system to be diagnosed,
COMPONENTS is a set of the system components,
OBS is a set of the observations of the system inputs
Output: A set of all diagnoses for (SD,COMPONENTS,OBS $\cup$ $\{\Pi_1, \ldots, \Pi_n\}$) given a sequence of measurements $\Pi_1, \ldots, \Pi_n$

```
CS := { }    ;a set of minimal conflict sets
DS := { }    ;a set of diagnoses
EC := { }    ;a set of equivalence classes
HD := { }    ;a set of homogeneous diagnoses
TCS := { }   ;a set of minimal conflict sets
1.  While the termination criterion is not satisfied do
2.    TCS := { }
3.    insert a probe into the system to make a measurement Π
4.    If CS={ }
5.      Then compute minimal conflict sets derived from COMPONENTS
```

```
        as described in Section 5.2 by using the pruning rules
        (2a) and (2b)
6.      update the CS with the above results
7.      compute diagnoses based on CS and DS according to
          Theorem 3.13
8.      update the DS with the above results
9.  Else If some member of DS predicts ¬ Π
10.     Then compute equivalence classes based on CS as
              described in Section 5.1
11.       update the EC with the above results as described
            in Section 5.1
12.       compute homogeneous diagnoses based on DS
            and EC as described in Section 5.1
13.       update the HD with the above results
14.       While some member(s) of DS predict ¬ Π do
15.         select one member Δ_i of DS predicting ¬ Π
              which has not been considered
16.         compute minimal conflict sets derived from
              COMPONENTS − Δ_i as described in Section 5.2
17.         add the results to TCS
18.         Repeat
19.           select any one Δ_j of Δ_i's homogeneous
                diagnosis predicting ¬ Π
20.           If Δ_j intersects with some of minimal conflict
                sets derived from COMPONENTS − Δ_i
21.             Then compute minimal conflict sets
                      derived from COMPONENTS −
                      Δ_j which intersect with Δ_i as
                      described in Section 5.2
22.                 add the results to TCS
23.           endif
24.         until all Δ_i's homogeneous diagnoses are
                considered
25.       endwhile
26.       CS := TCS
27.       compute diagnoses based on CS and DS according
            to Theorem 3.13
28.       update the DS with the above results
29.     endif
30.   endif
31. endwhile
32. return DS
33. endproc
```

Example 5.3. (*The familiar circuit with the indicated inputs*). The results obtained in the execution of our procedure DIAGNOSE on this circuit are shown below.

(1) SD (as addressed in Example 2.8), COMPONENTS = $\{M_1, M_2, M_3, A_1, A_2\}$, OBS = $\{\text{in1}(M_1) = 3,\ \text{in2}(M_1) = 2,\ \text{in1}(M_2) = 3,\ \text{in2}(M_2) = 2,\ \text{in1}(M_3) = 3,\ \text{in2}(M_3) = 2\}$, CS = $\{\}$, DS = $\{\}$, EC = $\{\}$, HD = $\{\}$.

(2) $\Pi_1 = \{\text{out}(A_1) = 10\}$

Step 2. TCS = $\{\}$

Step 4. "If" statement is true

Step 5. $\langle M_1, M_2, A_1\rangle$ is the minimal conflict set derived from COMPONENTS

Step 6. CS = $\{\langle M_1, M_2, A_1\rangle\}$

Steps 7, 8. DS = $\{[M_1], [M_2], [A_1]\}$

(3) $\Pi_2 = \{\text{out}(A_2) = 12\}$

Step 2. TCS = $\{\}$

Step 9. $[M_2]$ predicts $\neg\ \Pi_2$; $[M_2]$ predicts $\text{out}(A_2) = 10$

Steps 10, 11. EC = $\{[\![M_1, M_2, A_1]\!]\}$

Steps 12, 13. HD = $\{\{[M_1], [M_2], [A_1]\}\}$

Step 15. select $[M_2]$

Step 16. $\langle M_1, M_3, A_1, A_2\rangle$ is the minimal conflict set derived from COMPONENTS $- [M_2]$

Step 17. TCS = $\{\langle M_1, M_3, A_1, A_2\rangle\}$

Step 26. CS = $\{\langle M_1, M_3, A_1, A_2\rangle\}$

Steps 27, 28. DS = $\{[M_1], [A_1], [M_2, M_3], [M_2, A_2]\}$

(4) $\Pi_3 = \{\text{out}(M_2) = 5\}$

Step 2. TCS = $\{\}$

Step 9. $[M_1]$ predicts $\neg\ \Pi_3$

Steps 10, 11. EC = $\{[\![M_1, A_1]\!], [\![M_2]\!], [\![M_3, A_2]\!]\}$

Steps 12, 13. HD = $\{\{[M_1], [A_1]\}, \{[M_2, M_3], [M_2, A_2]\}\}$

Step 15. select $[M_1]$

Step 16. $\langle M_2\rangle$ and $\langle M_3, A_2\rangle$ are the minimal conflict sets derived from COMPONENTS $- [M_1]$

Step 17. TCS = $\{\langle M_2\rangle, \langle M_3, A_2\rangle\}$

Step 19. select $[A_1]$

Step 20. "If" statement is not satisfied

Step 15. select $[M_2, M_3]$

Step 16. $\langle M_1, A_1\rangle$ is the minimal conflict set derived from COMPONENTS $- [M_2, M_3]$

Step 17. TCS = $\{\langle M_2\rangle, \langle M_3, A_2\rangle, \langle M_1, A_1\rangle\}$

Step 19. select $[M_2, A_2]$

Step 20. "If" statement is not satisfied

Step 26. CS = $\{\langle M_2\rangle, \langle M_3, A_2\rangle, \langle M_1, A_1\rangle\}$

Steps 27, 28. DS = $\{[M_1, M_2, M_3], [M_1, M_2, A_2], [M_2, M_3, A_1], [M_2, A_1, A_2]\}$

We make several remarks about our procedure DIAGNOSE:

Remark 1. This procedure is relatively independent of the particular logic representing the system being diagnosed. Although our development of a theory of measurement in diagnosis has relied upon first-order logic as the underlying representation language for system descriptions, a close inspection of the preceding definitions, theorems, and proofs reveals that very few special features of first-order logic were actually required in developing the theory. Therefore, the logical representation language may be generalized. Notice that Definition 3.2 of a conflict set resulting from a measurement, the subsequent results of Section 3 leading to the procedure DIAGNOSE of Section 5.3, and the results of Section 4 on equivalence classes and homogeneous diagnoses require very weak assumptions on the nature of the logic. Specifically, suppose that L is any logic with the following properties:

(a) Its semantics is binary, namely every sentence of L has value true or false in a given structure.

(b) L has $\{\wedge, \vee, \neg\}$ among its logical connectives, and these have their usual interpretations.

Then we can generalize the concept of a system so that a system description and its observation can be any set of sentences of L. The definition of a conflict set resulting from a measurement remains the same in this generalized setting as in Definition 3.2. It is a simple matter to inspect the proofs of all results in Section 3 to see that they continue to hold, provided $\models$ is understood to denote the semantic entailment relation for the logic L. The procedure DIAGNOSE of Section 5.3 for calculating all diagnoses given a sequence of measurements remains the same and is correct for L. Of course, for DIAGNOSE really to be a procedure, we need a sound, complete and decidable theorem prover for L. Finally, inspection of the proofs of Section 4 reveals that all of our results on equivalence classes and homogeneous diagnoses continue to hold for L.

Therefore, our theory can apply equally well to a wide variety of logics, not just first-order. As a consequence, the procedure DIAGNOSE accommodates diagnostic reasoning in a wide variety of practical settings, including digital and analogue circuits, medicine, and database updates. For example, as Reiter addressed in [28], time varying digital hardware has natural representations in a temporal logic [19] and this might form the basis for a diagnostic reasoning system for such devices. Similarly, time varying physiological properties are central to certain kinds of medical diagnosis tasks [32]. Database logic has been proposed for representing many forms of database [16] so that violation of database integrity constraints may profitably be viewed as a diagnostic reasoning problem with database logic providing the system description language. These examples, and others like them, require a proper investigation with respect to problems of representation and computation. The fact that they all conform to a common theory of diagnosis is an encouraging and unifying observation.

As we remarked earlier, an underlying theorem prover which is used to determine whether a set of components is a conflict set is central to the performance of our procedure. We can use Reiter's TP function. But our TP function differs from his in that his TP returns a minimal conflict subset of some conflict set which is used as a parameter of a call to TP. Instead ours is only used to determine whether or not a set is a conflict. One way to realize the TP function computationally is to use a complete refutation based theorem prover which records the AB instances entering into the refutation it computes. In particular applications, TP might profitably be realized by special purpose theorem provers, e.g. boolean equation solvers for switching circuit, linear equation solvers for linear electronic circuit etc.

Therefore, such a special purpose theorem prover can be designed for particular diagnostic application, and can then "hook into" the general purpose procedure to yield a domain specific diagnostic procedure.

Remark 2. We do not calculate the diagnoses for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$) based on all minimal conflict sets for (SD,COMPONENTS,OBS $\cup$ $\{\Pi\}$). Instead, we calculate them based only on all minimal conflict sets resulting from Π. These minimal conflict sets are generated by computing the ones derived from COMPONENTS $-$ Δ in which Δ is an old diagnosis predicting $\neg\, \Pi$, instead of computing the minimal supporting environments of any value distinct from the measurement Π, as de Kleer and Williams mentioned in [7], from the bottom (namely the empty set ENV $= \{\}$) towards the root through a superset lattice which is developed in terms of

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg\, \text{AB}(c) \mid c \in \text{ENV}\}\ .$$

Remark 3. We do not compute all minimal conflict sets resulting from Π based on every COMPONENTS $-$ Δ determined by every old diagnosis Δ predicting $\neg\, \Pi$. Instead we compute the minimal conflict sets derived from COMPONENTS $-$ Δ_i and COMPONENTS $-$ Δ_j respectively in which Δ_i and Δ_j are not homogeneous. For the Δ_i and Δ_j which are homogeneous, if the ones derived from COMPONENTS $-$ Δ_i intersect with Δ_j, then we will compute the ones derived from COMPONENTS $-$ Δ_j which intersect with Δ_i. If not, then we need not consider the ones derived from COMPONENTS $-$ Δ_j.

Remark 4. For every previous diagnosis Δ predicting $\neg\, \Pi$, we compute all minimal conflict sets derived from COMPONENTS $-$ Δ by constructing a pruned conflict set tree rooted by COMPONENTS $-$ Δ depth-first. In the course of construction of the pruned CS-tree, two pruning enhancements (the pruning rules (2a) and (2b)) have been used in order to avoid unnecessary access to the theorem prover. Therefore, all different minimal conflicts resulting from Π can be straightforwardly generated and no two calls by DIAGNOSE to TP will ever return the same conflict set.

Remark 5. We do not calculate diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) based on old minimal conflicts for (SD,COMPONENTS,OBS). Instead, we calculate them based on the old diagnoses and new minimal conflicts resulting from Π. Furthermore, we determine homogeneous diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) that are also based on the minimal conflicts resulting from Π.

Remark 6. Our procedure is incremental, exploiting the iterative nature of diagnosis. It calculates not only the diagnoses for (SD,COMPONENTS,OBS) with a fixed observation OBS but also the ones for (SD,COMPONENTS,OBS $\cup$ $\{\Pi_1, \ldots, \Pi_n\}$) with a sequence of measurements $\Pi_1, \ldots, \Pi_n$.

6. Pragmatics

6.1. Completeness and most probable candidates

By Theorems 3.6 and 3.13, we know that our conflict recognition strategy is complete, resulting in the underlying inference process being complete. As a consequence, in our approach, not only do the old diagnoses which predict Π (a measurement value) remain, but the new diagnoses which are the strict superset of some old diagnosis predicting $\neg\, \Pi$ will be generated.

In Example 2.8 above, for instance, we have four old diagnoses: $[M_1]$, $[A_1]$, $[M_2, M_3]$ and $[M_2, A_2]$. Given a measurement $\Pi = \{\text{out}(M_3) = 8\}$, all minimal conflict sets resulting from Π are $\langle M_3 \rangle$ and $\langle M_2, A_2 \rangle$. By Theorem 3.13, we determine all new candidates: $[M_2, M_3]$, $[M_1, M_3, A_2]$ and $[M_3, A_1, A_2]$. Here, the old diagnosis $[M_2, M_3]$ remains while the new diagnoses $[M_1, M_3, A_2]$ and $[M_3, A_1, A_2]$ are generated.

In practice, computing all new candidates is computationally expensive. Therefore, it might be necessary to compute the more probable candidates. There is no way to tell whether a single candidate actually has a high probability without knowing the overall normalization factors. However, in most cases, the remaining previous diagnoses may be very probable, since it is confirmed by the measurement.

In the example above, given a measurement $\Pi = \{\text{out}(M_3) = 8\}$, we can keep the old diagnosis $[M_2, M_3]$ without computing the two new diagnoses $[M_1, M_3, A_2]$ and $[M_3, A_1, A_2]$.

Therefore we might revise the procedure DIAGNOSE by modifying Step 9 into Steps 9a and 9b as follows:

9a. **Else If** some member of DS predicts Π
9b. **Then** update the DS by removing all previous diagnoses predicting $\neg\, \Pi$

10. **Else** compute equivalence classes based on CS as described in Section 5.1

As we said earlier, this revised procedure might be incomplete. If the remaining candidates are not the actual bugs, we should compute the new candidates which are the superset of some previous diagnoses predicting $\neg \Pi$.

6.2. When to stop diagnosing

In Step 1 of the procedure DIAGNOSE, the termination criterion should be tested. If we did not design a reasonable criterion, the procedure would continue forever. Of course, this is impractical. On the other hand, we notice that a special criterion can be designed for particular diagnostic application.

In this section we discuss some general factors affecting the design of termination criterion.

Criterion 1. If we obtain only one diagnosis for (SD,COMPONENTS,OBS $\cup$ $\{\Pi_1, \Pi_2, \ldots, \Pi_n\}$) after making a sequence of measurements $\Pi_1, \ldots, \Pi_n$, we can stop the procedure. This is because the objective of making measurements is to discriminate among competing diagnoses. If there is only one diagnosis, no further discrimination via measurements is needed.

Criterion 2. Because diagnostic reasoning is nonmonotonic, especially in the sense that it can happen that none of a system's diagnoses survive a new observation of that system, a measurement probably increases the size of the candidate space, not merely discriminating among competing diagnoses. Therefore, for systems with large numbers of components, computing new diagnoses given a new measurement might increase the computational cost. A good heuristic to reduce the computational cost is to set some threshold fraction, such as some small number that we are willing to consider as the upper bound on the acceptable size of the set of final diagnoses produced. Our procedure can be stopped for some candidates computed below the threshold fraction. But, if the resultant candidates are not the actual ones, the procedure should be rerun until the actual candidates are computed.

Criterion 3. By Theorem 2.11, we notice that a diagnosis, once eliminated, can never reappear. Therefore, if we once generate a diagnosis of cardinality K, we can never calculate any diagnoses of cardinality less than K which are the subsets of that diagnosis. This means that, in general, the diagnoses' cardinality will increase as our procedure calculates all diagnoses given a sequence of measurements. Thus, if, for some reason, we believe that diagnoses of cardinality greater than k are highly improbable, or if we are interested only in diagnoses of cardinality k or less, we can stop the procedure while generating all diagnoses of cardinality k.

Criterion 4. If no measurement points remain in the system, then no further information can be obtained. DIAGNOSE should be stopped.

7. Relationship to other research

The primary influences on this paper have been the work of Reiter [28] and de Kleer and Williams [7].

7.1. Reiter

Reiter explored many of the ideas for diagnosis. His theory of diagnosis provides a formal analysis of multiple fault diagnoses. Some of his results (e.g. Theorem 2.11) provide a basis for our theory. However, his theory does not include a theory of measurements. Although he proves various results about the effects of system measurements on diagnoses, he does not indicate how to compute new diagnoses for (SD,COMPONENTS,OBS $\cup \{\Pi\}$) based on the old diagnoses for (SD,COMPONENTS,OBS). Hence one cannot compute new diagnoses using Reiter's algorithm unless he calls (SD,COMPONENTS,OBS $\cup \{\Pi\}$) again. As we discussed above, the computation of (SD,COMPONENTS,OBS $\cup \{\Pi\}$) will repeat the work of (SD,COMPONENTS,OBS).

Our theory proposes an approach to computing all new diagnoses resulting from a new measurement based upon the old diagnoses. Our procedure first computes all minimal conflict sets resulting from a new measurement Π derived from every COMPONENTS $- \Delta$ in which Δ is an old diagnosis predicting $\neg \Pi$. Then our procedure generates all new diagnoses based upon the new minimal conflicts and the old diagnoses, without repeating the computation of DIAGNOSE(SD,COMPONENTS,OBS). Therefore, whenever we make a new measurement, we can continue our procedure until we obtain a satisfactory result.

7.2. De Kleer and Williams

In their 1987 paper, de Kleer and Williams independently proposed a characterization of diagnoses and an approach, namely the GDE, to computing all new diagnoses resulting from a new measurement. Their work differs from ours, however, in lacking a formalization of their diagnostic theory. As we remarked in Section 3, our Theorem 3.15 can be interpreted as a formal justification for their GDE's procedure. On the other hand, de Kleer and Williams go beyond our theory of measurement in diagnosis by providing a method for computing the probabilities of the different diagnoses that arise in a given setting, and for using these probabilities to identify the appropriate system measurement to make next.

As GDE's procedure does, our procedure should compute all minimal conflict sets resulting from a new measurement first. Our method differs from theirs, however. De Kleer and Williams calculate these minimal conflicts by computing the minimal supporting environments of any value distinct from the measurement Π from the bottom (namely the empty set ENV $= \{\}$) towards the root through a superset lattice which is developed breadth-first in terms of

$$\text{SD} \cup \text{OBS} \cup \{\Pi\} \cup \{\neg\, \text{AB}(c) \mid c \in \text{ENV}\}\ ,$$

while we determine them by computing the minimal conflicts derived from COMPONENTS $- \Delta$, in which Δ is an old diagnosis predicting $\neg\, \Pi$, from the root (namely the set COMPONENTS $- \Delta$) towards the bottom through a pruned CS-tree which is developed depth-first. By using the pruning devices described in Sections 5.1 and 5.2, our procedure will require less computation than GDE's architecture demands for systems with large numbers of components.

In their 1989 paper [8], de Kleer and Williams have developed a new diagnostic procedure Sherlock—an extended GDE by introducing the behavioral modes per component. A component's modes consist of a set describing the component's proper behavior (e.g. the value being on or off), a set describing faulty behavior (e.g. the value being clogged or leaky) and an unknown mode. By using the methods for behavioral modes together with probabilistic information about the likelihood of each mode of behavior, they enhance their procedure's efficiency through exacerbating the process of diagnosing multiple faults and the process of discriminating among competing candidates.

7.3. Genesereth and Davis

Both Genesereth [9] and Davis [3] have addressed system test. Genesereth uses the first-order logic for representing system and proposes a resolution style theorem prover for computing candidate faults as well as for generating tests. Davis proposes an approach to diagnosis from first principles, but oriented towards devices simulatable by constraint propagation techniques. Their method for generating a test is to choose a new set of system inputs, predict an observable proposition and compute new suspect components. They use the new candidates to confirm or disconfirm the old ones so as to discriminate among competing diagnoses. Obviously, their theory differs from ours and comparisons between the two methods are difficult to make. However, their approach is useful in cases where it is both difficult to make internal measurements and easy to change system inputs.

In their approach, if some of the new candidates derived from the new test has some overlap with the old ones, namely, the new one accounts for all the observed behavior, then that test is successful and the competing diagnoses can be discriminated. If neither of the new candidates are found in the set of the old candidates, then no components are capable of explaining the data from both sets of inputs, and another kind of hypothesis (e.g. bridge faults) should be considered. Davis has addressed the bridge faults by surrendering the assumption that the structure is as shown in the schematic and considering one class of modifications to the structure namely the addition of one wire between physically adjacent pins. His analysis of bridge faults in digital circuits is beyond the capabilities of our theory because our theory requires a fixed, a priori enumeration of the system components which might fail.

7.4. Others

There are two other approaches to diagnosis from first principles. One approach is by David Poole and his colleagues [17, 20] who have observed the connection of diagnostic reasoning with default logic. Both references describe how a default logic theorem prover can be used to compute diagnoses, but the focus of these papers is merely on mechanisms for such computations. The other logically based approach to diagnosis is by Ginsberg [10] who adopts a logic of counterfactual implication as a foundation for diagnostic reasoning. Following Genesereth [9], Ginsberg assumes a first-order representation of the system being diagnosed. But, his departure from Genesereth is to define diagnosis in terms of counterfactual consequences of the system observation.

A recent theory of diagnosis is the GSC (generalized set covering) model of Reggia, Nau and Wang [24, 26]. This provides a formal model of what they call "abductive diagnostic inference" and has been applied to problems of medical diagnosis [24]. They define a diagnosis (an explanation) by a minimum cardinality cover of M^+, a set of manifestations which have been observed to occur in the current diagnostic setting. In fact, the diagnosis they defined is a minimal hitting set for the collection of all minimal conflict sets (disorders).

Exploiting the use of fault models in model-based diagnosis has recently become an active research area [12, 15, 22, 31], leading to a direction of informed generate-and-test architectures. The incorporation of failure modes with the model-based diagnosis is to keep the computational explosion, whether in the process of diagnosing multiple faults or in the process of discriminating competing diagnoses, under control. As de Kleer and Williams addressed in [8], Holtzblatt [15] developed the GMODS system by incorporating the notion of behavioral modes with his generalization of GDE. GMODS relies on an expensive hyper-resolution rule to rule out fault modes so as to generate candidates. Struss [31] employs resolution rules and controls reasoning to introduce appropriate fault modes only when necessary. Through the use of an alternative architecture which redefines the notion of fault, Raimon [22] achieves some of the advantage of knowledge of fault modes without having to incorporate them. Hamscher [12] developed XDE by incorporating fault models with his generalization of GDE.

8. Summary

We summarize what we take to be the main contributions of this paper to a theory of measurement in diagnosis from first principles:

(1) The definition of the concept of equivalence classes and homogeneous diagnoses.

(2) The method for calculating all minimal conflict sets resulting from a measurement Π based upon constructing a pruned CS-tree rooted by the component set COMPONENTS $- \Delta$ depth-first, in which Δ is a previous diagnosis predicting $\neg \Pi$.
(3) The technique for determining new equivalence classes and new homogeneous diagnoses based upon all minimal conflict sets resulting from Π.
(4) The efficient incremental method for computing new diagnoses given a new measurement based upon the previous diagnosis predicting the opposite.
(5) The ability of the theory to accommodate a wide variety of logics.
(6) The procedure DIAGNOSE for calculating all the diagnoses resulting from a sequence of measurements $\Pi_1, \ldots, \Pi_n$.

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